

notebook

September 23, 2025

1 To do

- ~~Analyze lick data~~
 - ~~RT~~
 - ~~Lick rate~~
 - ~~ILI~~
- History kernels
- ~~Add HMM~~
- Repeating bias per ILD

2 Analyses of drug data from batch #6

3 Load behavioral data

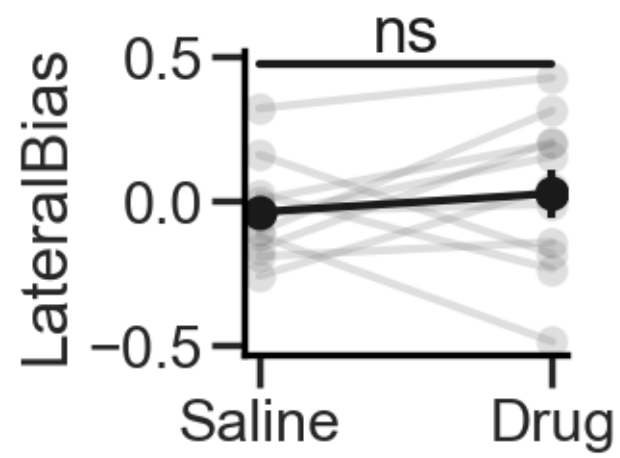
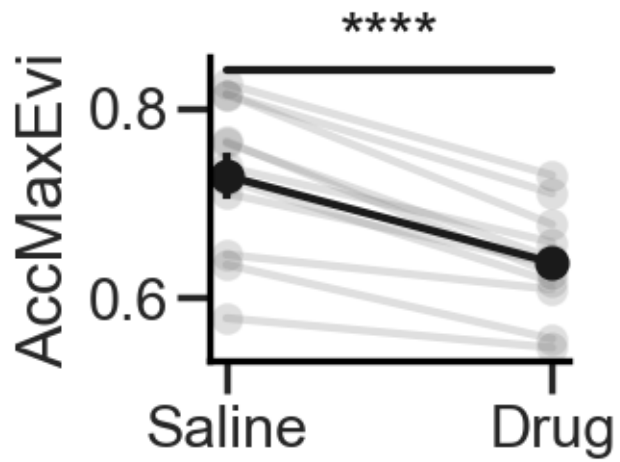
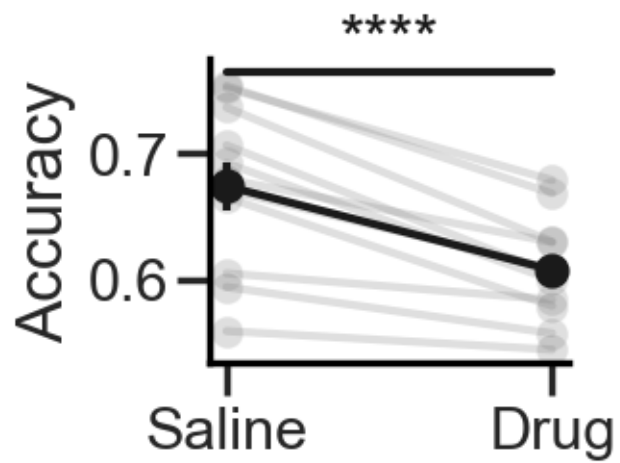
4 Filter data

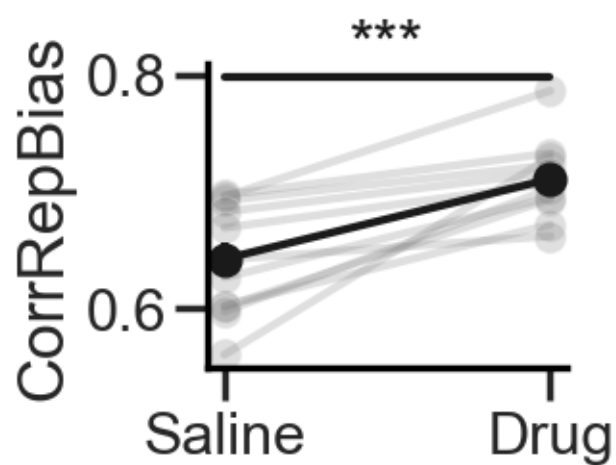
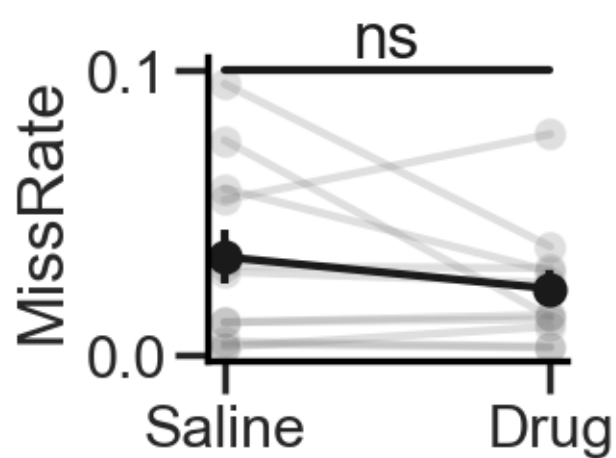
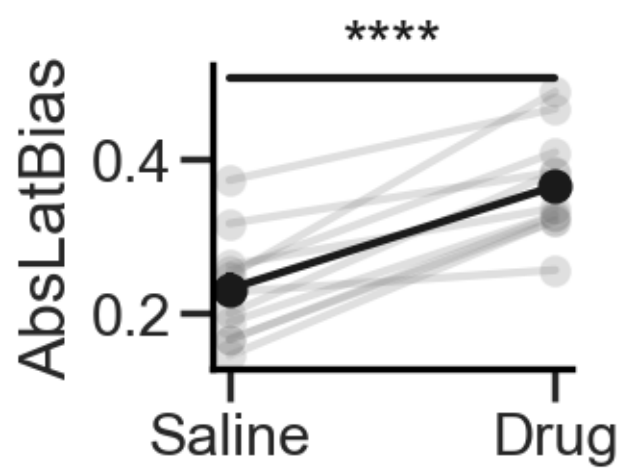
```
91 paired sessions
Bad sessions: 64      32
141      70
143      71
145      72
162      81
181      90
Name: PairedSession, dtype: int64
Dropped 6.593406593406594% paired sessions
```

5 Behavioral metrics

5.1 Standard

```
Accuracy: t-statistic = 6.877, p-value = 0.000
AccMaxEvi: t-statistic = 8.415, p-value = 0.000
LateralBias: t-statistic = -0.761, p-value = 0.464
AbsLatBias: t-statistic = -7.229, p-value = 0.000
MissRate: t-statistic = 1.368, p-value = 0.201
CorrRepBias: t-statistic = -5.146, p-value = 0.000
```





5.2 Licks

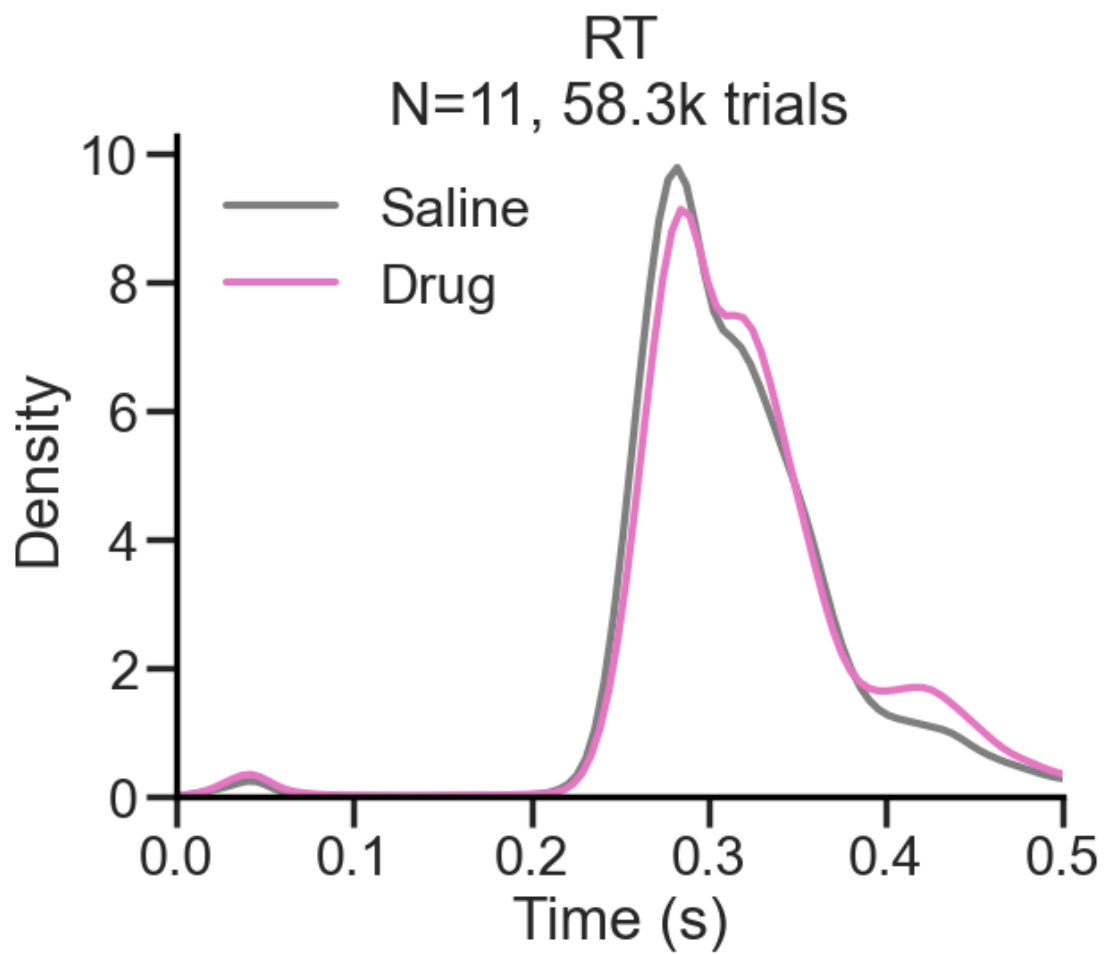
5.2.1 Add lick data

3.27% of premature lick trials (before response window)

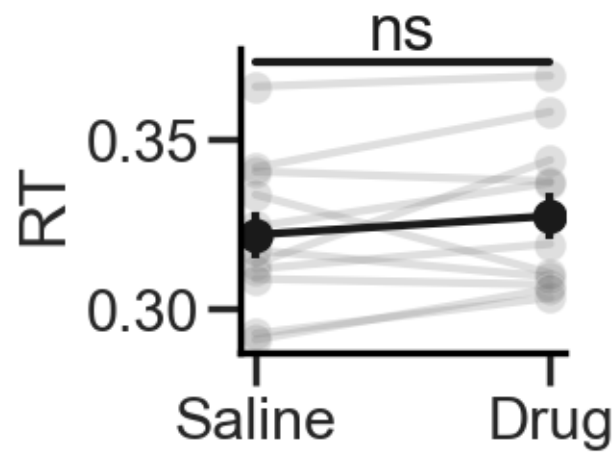
Port1In or Port2In are already lists

```
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',  
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

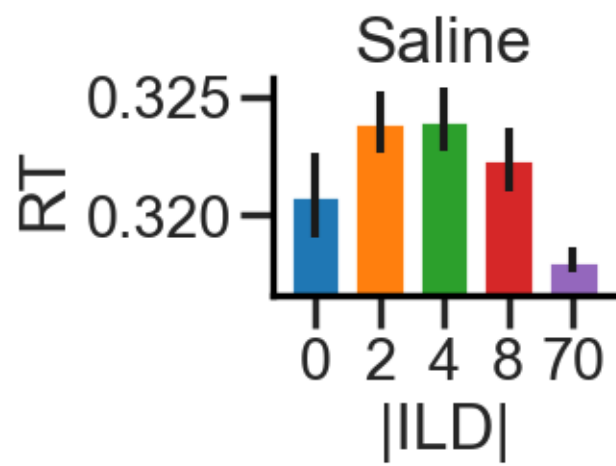
5.2.2 RTs



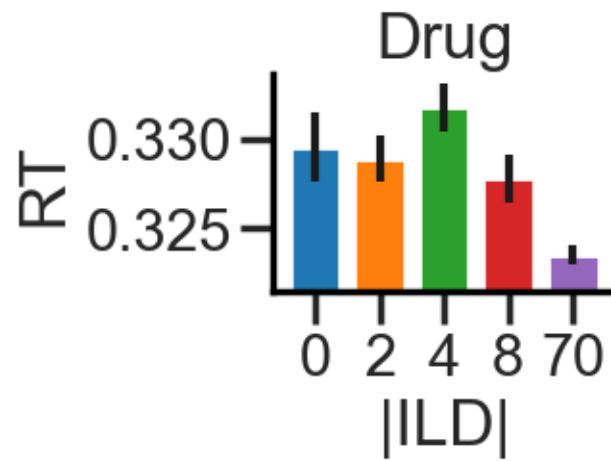
RT: t-statistic = -1.269, p-value = 0.233



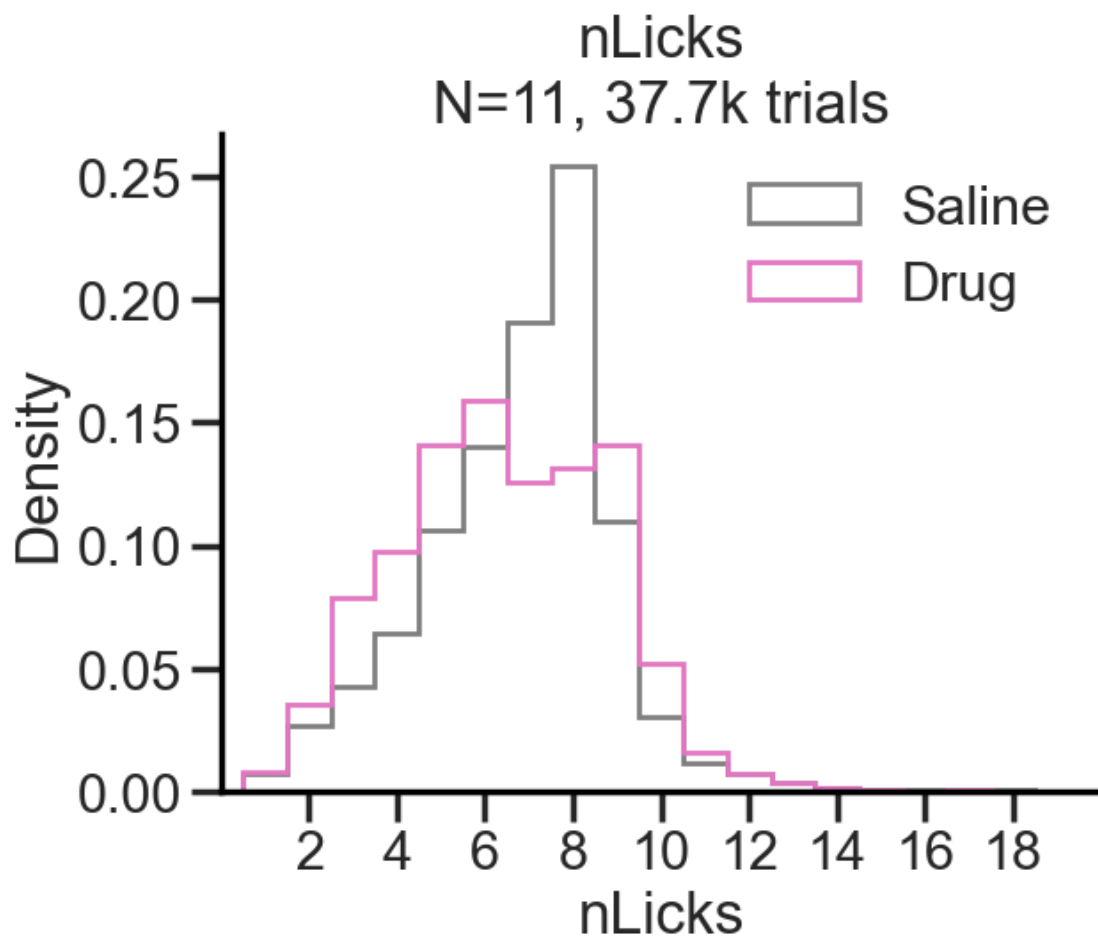
Text(0.5, 1.0, 'Saline')



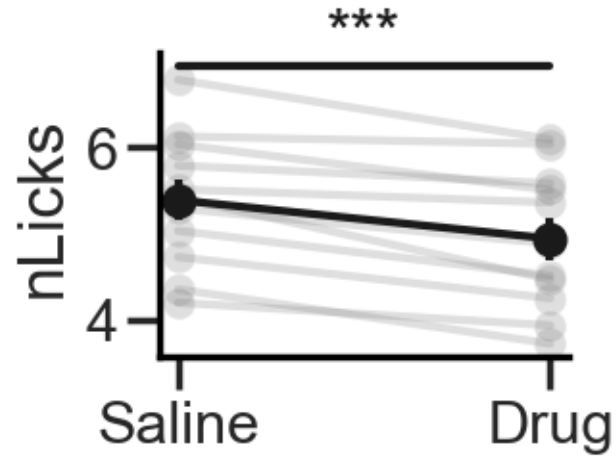
Text(0.5, 1.0, 'Drug')



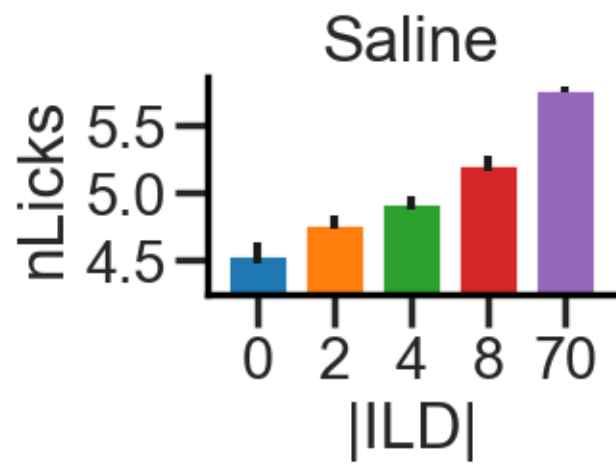
5.2.3 Lick rate



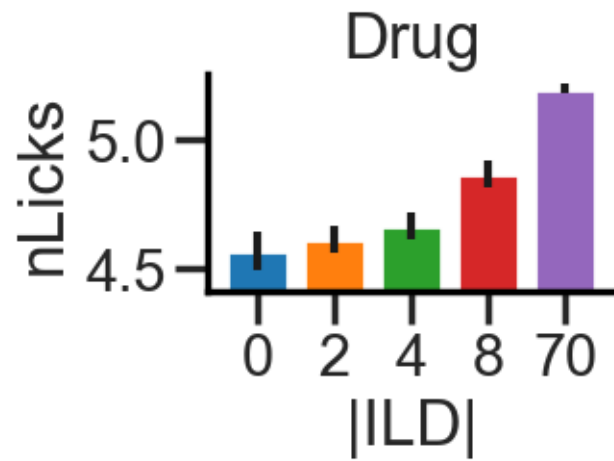
nLicks: t-statistic = 5.736, p-value = 0.000



Text(0.5, 1.0, 'Saline')

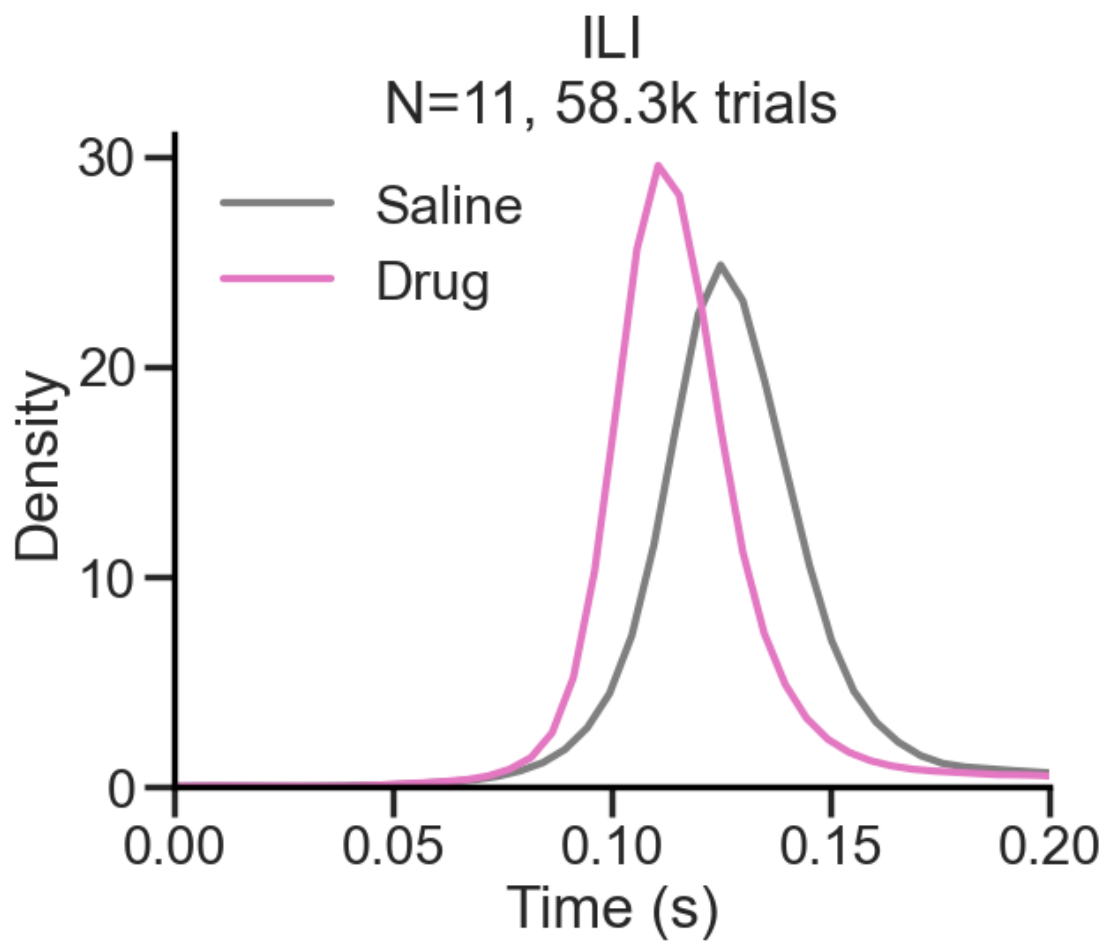


Text(0.5, 1.0, 'Drug')

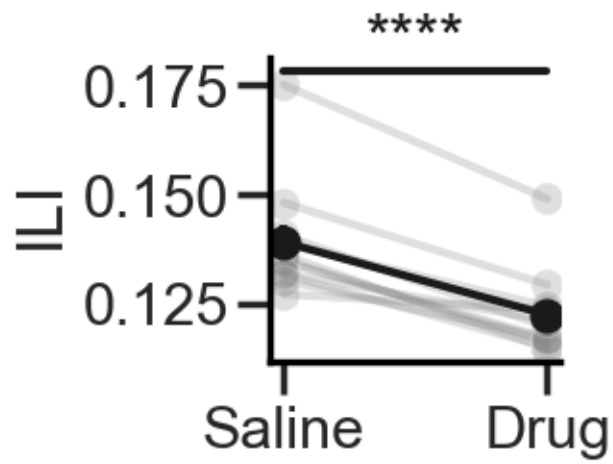


5.2.4 ILI

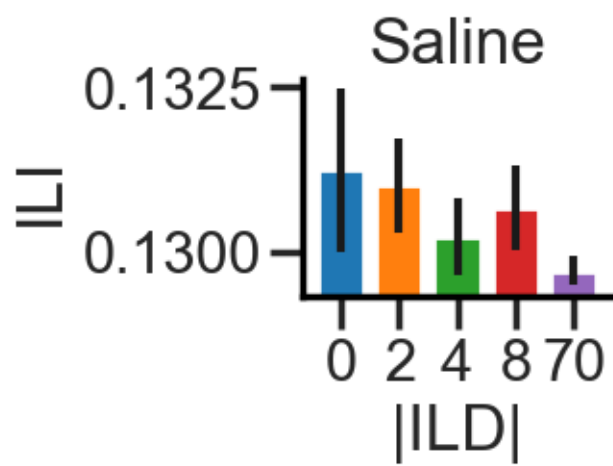
(0.0, 0.2)



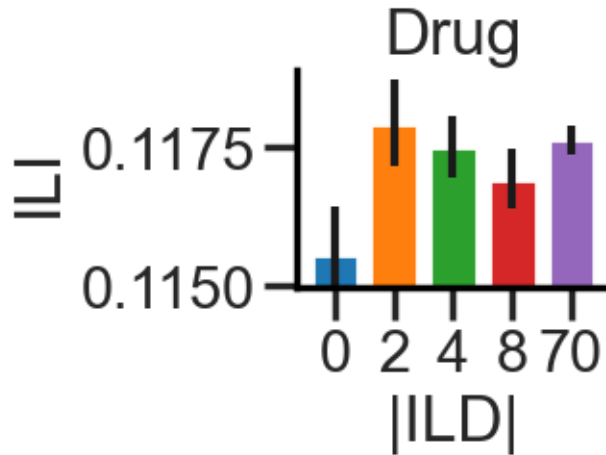
ILI: t-statistic = 9.096, p-value = 0.000



Text(0.5, 1.0, 'Saline')



Text(0.5, 1.0, 'Drug')



Mixed Effects Model To show that the drug removes the ILD dependence you have to show that the interaction is significant. Do a Linear Mixed model of the ILI1 with norm_trial_index, norm_trial_index², ILD, drug, drug x ILD, current_outcome, prev_outcome, Left-Right response (binary variable to account for differences in the port side). Put session as the random effect. Do it for each mouse and plot the stats of the weights. Let's see what comes out significant. You can use the same design matrix for the Nicks and the z_transformed_RT.

```
C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
```

```
warnings.warn(msg, ConvergenceWarning)
```

```
C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
```

```
warnings.warn(msg, ConvergenceWarning)
```

```
C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
```

```
warnings.warn(msg, ConvergenceWarning)
```

Mixed Linear Model Regression Results

```
=====
```

```
Model:                MixedLM Dependent Variable: ILI
No. Observations: 4017   Method:                REML
No. Groups: 14          Scale:                0.0013
Min. group size: 138     Log-Likelihood:    7672.9873
Max. group size: 324     Converged:         Yes
Mean group size: 286.9
```

```
-----
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
```

Intercept	0.138	0.003	44.048	0.000	0.132	0.144
absILD	0.000	0.000	2.459	0.014	0.000	0.000
Drug	-0.019	0.004	-4.356	0.000	-0.028	-0.011
Drug:absILD	-0.000	0.000	-1.594	0.111	-0.000	0.000
Group Var	0.000	0.001				

=====

Mixed Linear Model Regression Results

```

=====
Model:                MixedLM Dependent Variable: ILI
No. Observations: 5642   Method:                REML
No. Groups:           18   Scale:                0.0016
Min. group size:     227   Log-Likelihood:    10087.4440
Max. group size:     334   Converged:         Yes
Mean group size:     313.4

```

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.147	0.002	73.488	0.000	0.143	0.151
absILD	0.000	0.000	0.784	0.433	-0.000	0.000
Drug	-0.020	0.003	-7.004	0.000	-0.025	-0.014
Drug:absILD	0.000	0.000	0.414	0.679	-0.000	0.000
Group Var	0.000	0.000				

=====

Mixed Linear Model Regression Results

```

=====
Model:                MixedLM Dependent Variable: ILI
No. Observations: 6334   Method:                REML
No. Groups:           20   Scale:                0.0008
Min. group size:     298   Log-Likelihood:    13467.3301
Max. group size:     332   Converged:         Yes
Mean group size:     316.7

```

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.133	0.002	78.801	0.000	0.130	0.136
absILD	-0.000	0.000	-1.808	0.071	-0.000	0.000
Drug	-0.019	0.002	-8.069	0.000	-0.024	-0.015
Drug:absILD	0.000	0.000	2.274	0.023	0.000	0.000
Group Var	0.000	0.000				

=====

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning: The MLE may be on the boundary of the parameter space.
warnings.warn(msg, ConvergenceWarning)

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.

warnings.warn(msg, ConvergenceWarning)

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.

warnings.warn(msg, ConvergenceWarning)

Mixed Linear Model Regression Results

```
=====
Model:                MixedLM Dependent Variable: ILI
No. Observations: 2540    Method:                REML
No. Groups:           10    Scale:                0.0010
Min. group size:      212    Log-Likelihood:      5099.1975
Max. group size:      306    Converged:           Yes
Mean group size:      254.0

-----
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
Intercept      0.135    0.003 50.444 0.000  0.130  0.140
absILD         0.000    0.000  0.385 0.700 -0.000  0.000
Drug          -0.024    0.004 -6.550 0.000 -0.031 -0.017
Drug:absILD    0.000    0.000  1.682 0.093 -0.000  0.000
Group Var      0.000    0.000
=====
```

Mixed Linear Model Regression Results

```
=====
Model:                MixedLM Dependent Variable: ILI
No. Observations: 2243    Method:                REML
No. Groups:           8    Scale:                0.0039
Min. group size:      248    Log-Likelihood:      2992.6927
Max. group size:      307    Converged:           Yes
Mean group size:      280.4

-----
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
Intercept      0.179    0.006 31.453 0.000  0.168  0.190
absILD        -0.000    0.000 -1.341 0.180 -0.000  0.000
Drug          -0.036    0.008 -4.455 0.000 -0.051 -0.020
Drug:absILD    0.000    0.000  2.408 0.016  0.000  0.000
Group Var      0.000    0.001
=====
```

Mixed Linear Model Regression Results

```
=====
Model:                MixedLM Dependent Variable: ILI
```

```

No. Observations: 6972    Method:          REML
No. Groups:          22    Scale:           0.0009
Min. group size:    283    Log-Likelihood: 14699.9363
Max. group size:    350    Converged:        Yes
Mean group size:    316.9

```

```

-----
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
Intercept      0.134    0.002 83.251 0.000   0.131   0.138
absILD        -0.000    0.000 -0.812 0.417  -0.000   0.000
Drug          -0.020    0.002 -8.571 0.000  -0.024  -0.015
Drug:absILD     0.000    0.000  2.317 0.020   0.000   0.000
Group Var       0.000    0.000
=====

```

```

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)

```

Mixed Linear Model Regression Results

```

=====
Model:              MixedLM Dependent Variable: ILI
No. Observations: 7057    Method:          REML
No. Groups:          23    Scale:           0.0014
Min. group size:    116    Log-Likelihood: 13116.0827
Max. group size:    338    Converged:        Yes
Mean group size:    306.8

```

```

-----
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
Intercept      0.129    0.002 52.534 0.000   0.124   0.134
absILD        -0.000    0.000 -1.482 0.138  -0.000   0.000
Drug          -0.008    0.004 -2.310 0.021  -0.015  -0.001
Drug:absILD     0.000    0.000  2.918 0.004   0.000   0.000
Group Var       0.000    0.001
=====

```

Mixed Linear Model Regression Results

```

=====
Model:              MixedLM Dependent Variable: ILI
No. Observations: 4687    Method:          REML
No. Groups:          19    Scale:           0.0011
Min. group size:    18    Log-Likelihood: 9304.2857

```

Max. group size: 328 Converged: Yes
Mean group size: 246.7

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.138	0.004	30.588	0.000	0.129	0.146
absILD	-0.000	0.000	-1.097	0.273	-0.000	0.000
Drug	-0.019	0.006	-2.985	0.003	-0.032	-0.007
Drug:absILD	0.000	0.000	1.873	0.061	-0.000	0.000
Group Var	0.000	0.002				

Mixed Linear Model Regression Results

Model: MixedLM Dependent Variable: ILI
No. Observations: 4676 Method: REML
No. Groups: 18 Scale: 0.0013
Min. group size: 213 Log-Likelihood: 8843.0451
Max. group size: 304 Converged: Yes
Mean group size: 259.8

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.129	0.002	57.519	0.000	0.125	0.134
absILD	0.000	0.000	0.507	0.612	-0.000	0.000
Drug	-0.013	0.003	-4.022	0.000	-0.019	-0.007
Drug:absILD	-0.000	0.000	-0.212	0.832	-0.000	0.000
Group Var	0.000	0.000				

Mixed Linear Model Regression Results

Model: MixedLM Dependent Variable: ILI
No. Observations: 4876 Method: REML
No. Groups: 18 Scale: 0.0016
Min. group size: 146 Log-Likelihood: 8775.7966
Max. group size: 325 Converged: Yes
Mean group size: 270.9

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.131	0.006	23.461	0.000	0.120	0.142
absILD	0.000	0.000	0.276	0.783	-0.000	0.000
Drug	-0.012	0.008	-1.552	0.121	-0.028	0.003
Drug:absILD	0.000	0.000	1.301	0.193	-0.000	0.000
Group Var	0.000	0.002				

Mixed Linear Model Regression Results

```

=====
Model:                MixedLM Dependent Variable: ILI
No. Observations: 4828   Method:                REML
No. Groups:           18   Scale:                0.0029
Min. group size:     131   Log-Likelihood:    7203.7654
Max. group size:     326   Converged:         Yes
Mean group size:     268.2
=====

```

```

-----
              Coef.  Std.Err.   z    P>|z|  [0.025 0.975]
-----
Intercept      0.145      0.004 35.852 0.000   0.137   0.153
absILD        -0.000      0.000 -3.910 0.000  -0.000  -0.000
Drug          -0.024      0.006 -4.138 0.000  -0.035  -0.012
Drug:absILD     0.000      0.000  4.519 0.000   0.000   0.000
Group Var       0.000      0.001
=====

```

```

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.

```

```

    warnings.warn(msg, ConvergenceWarning)

```

```

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.

```

```

    warnings.warn(msg, ConvergenceWarning)

```

```

C:\Users\Usuario\anaconda3\envs\analysis\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.

```

```

    warnings.warn(msg, ConvergenceWarning)

```

```

Intercept      0.139888

```

```

absILD        -0.000018

```

```

Drug          -0.019384

```

```

Drug:absILD     0.000068

```

```

dtype: float64

```

```

Intercept      0.004279

```

```

absILD         0.000017

```

```

Drug           0.002180

```

```

Drug:absILD     0.000025

```

```

dtype: float64

```

```

Intercept      9.263232e-123

```

```

absILD         3.291207e-01

```

```

Drug           1.312816e-02

```

```

Drug:absILD     1.847074e-01

```

```

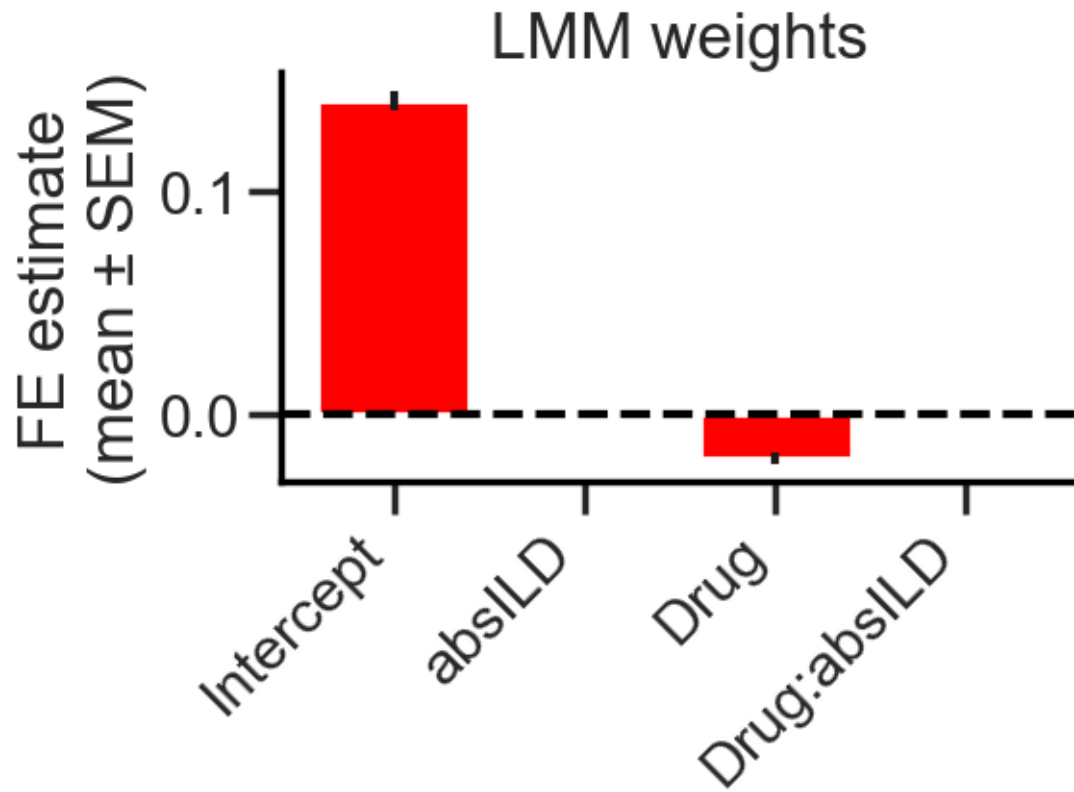
Group Var      3.330676e-02

```

```

dtype: float64

```

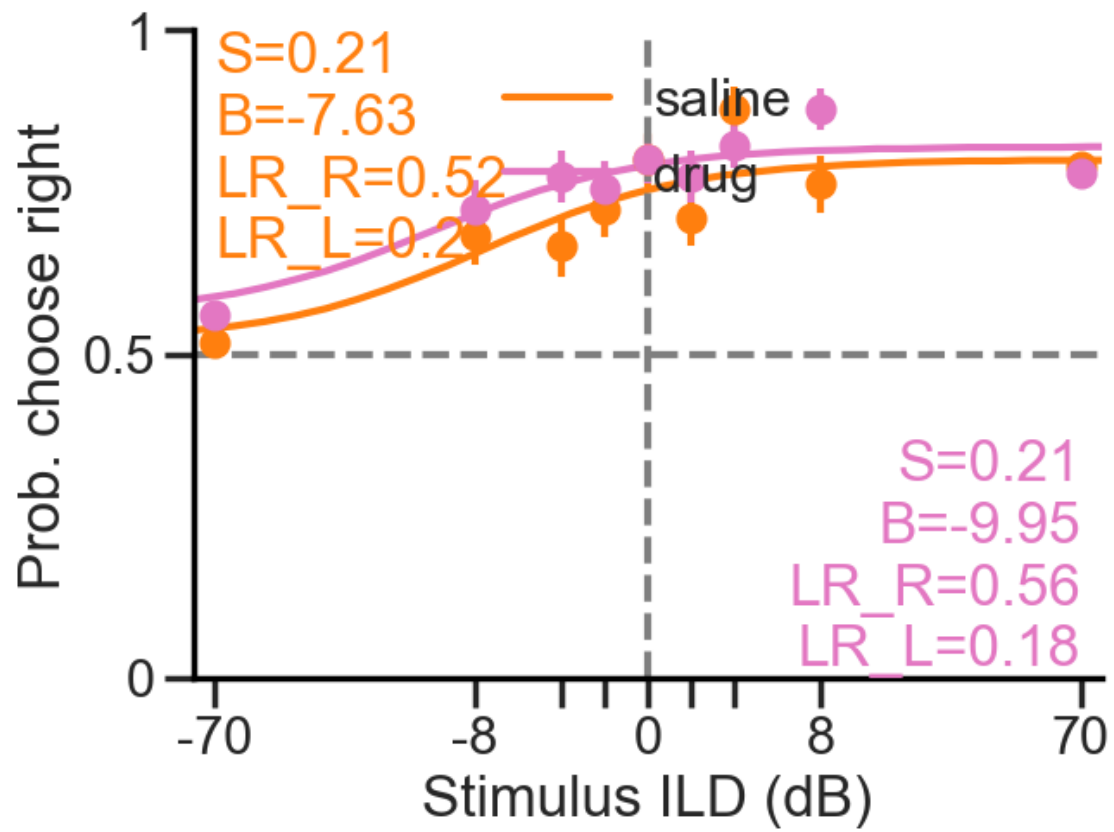


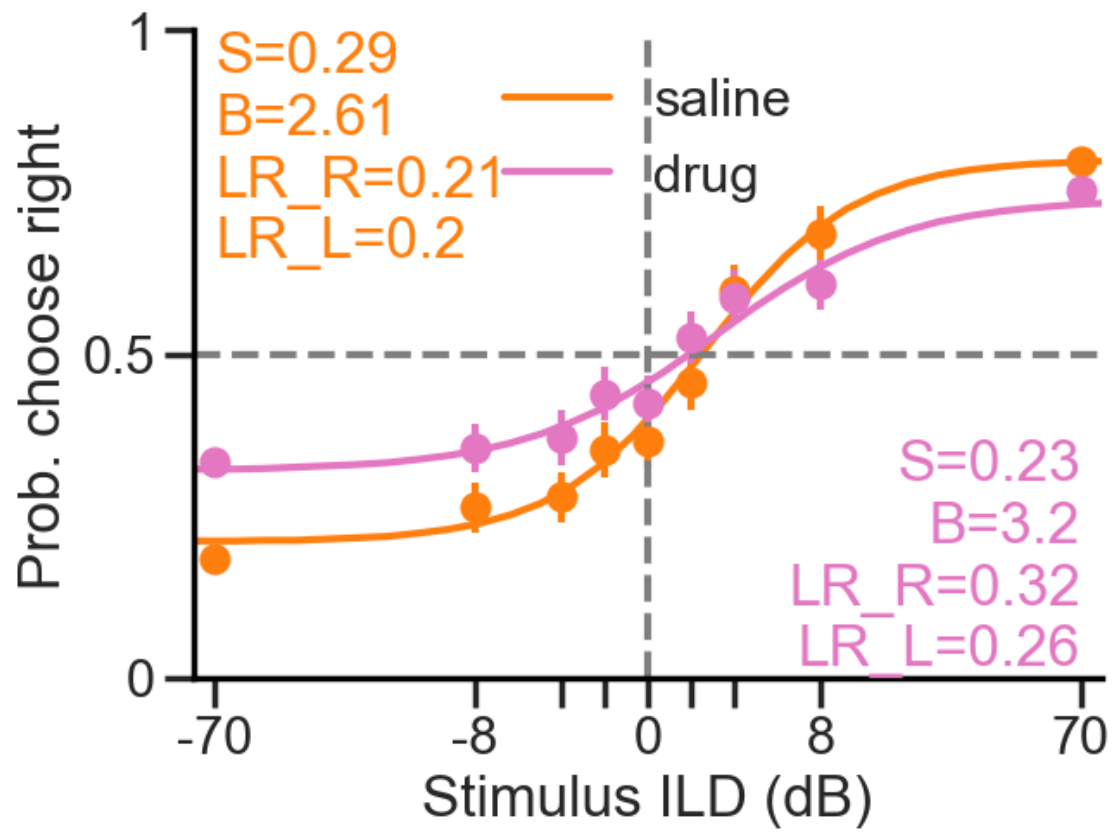
6 Psychometrics

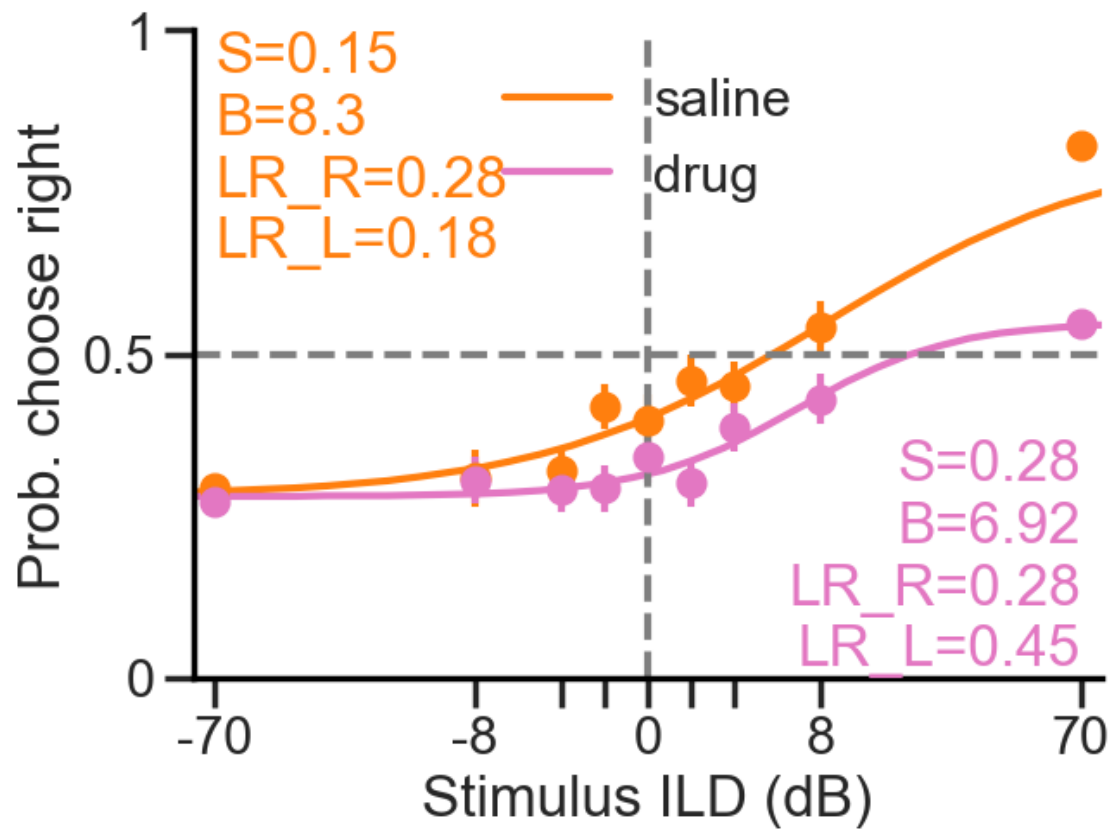
6.1 Probability choose RIGHT

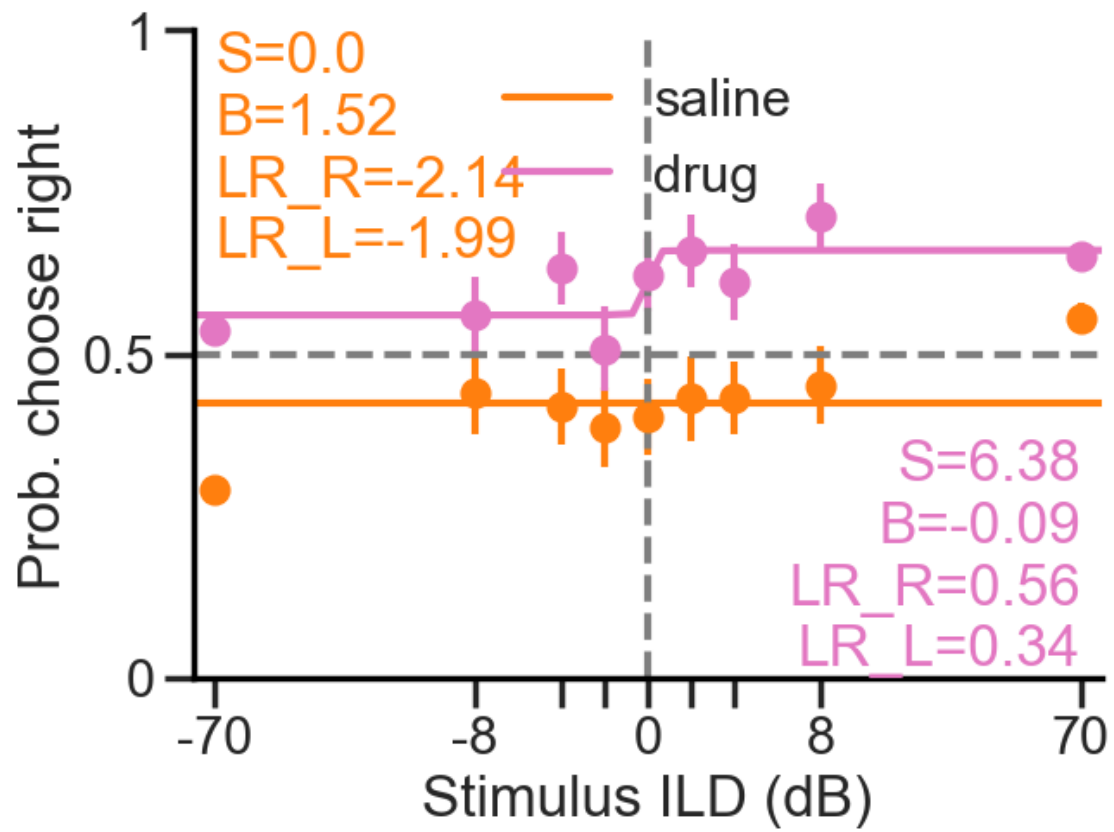
6.1.1 Single animals

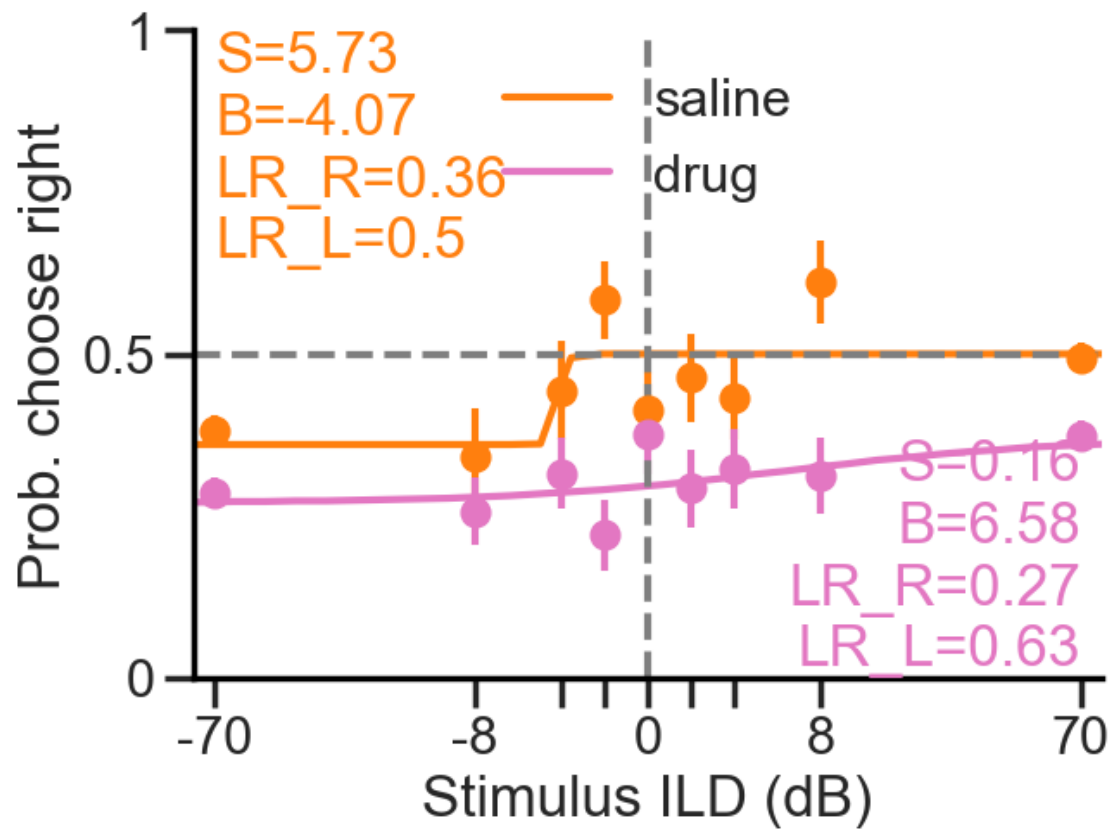
Animal 014
 Animal 016
 Animal 017
 Animal 018
 Animal 019
 Animal 020
 Animal 021
 Animal 022
 Animal 023
 Animal 024
 Animal 025

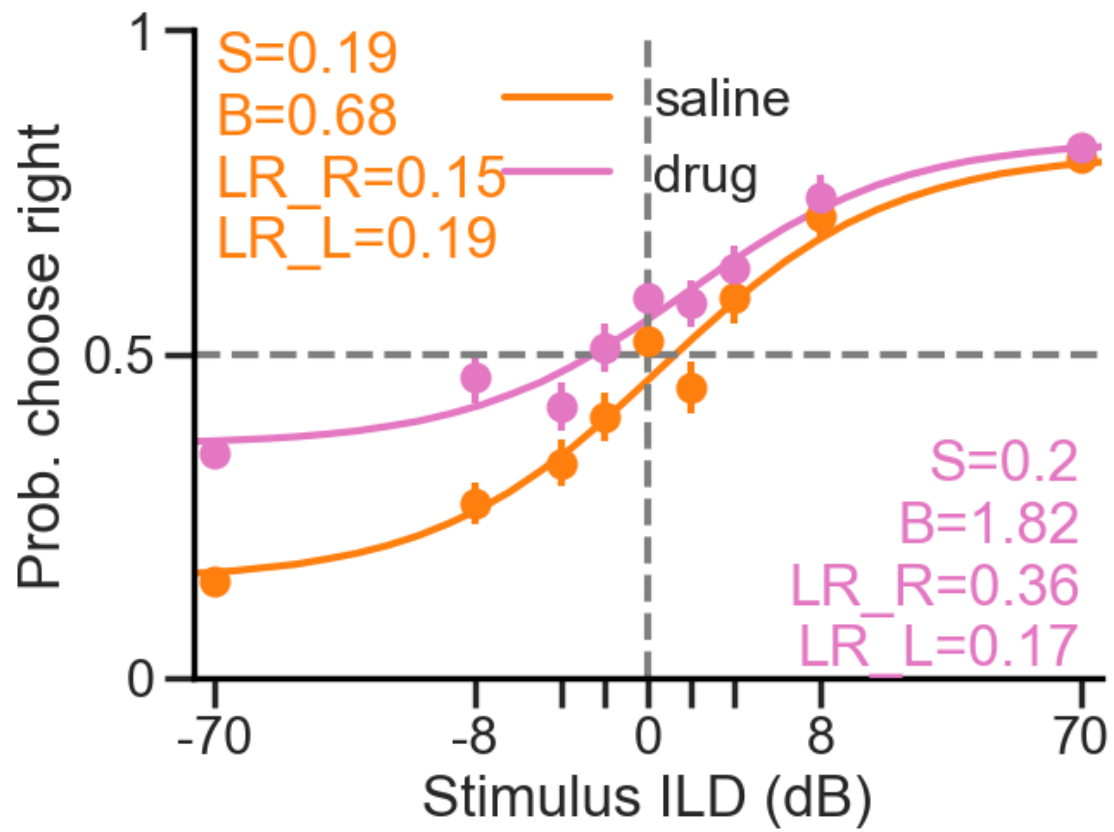


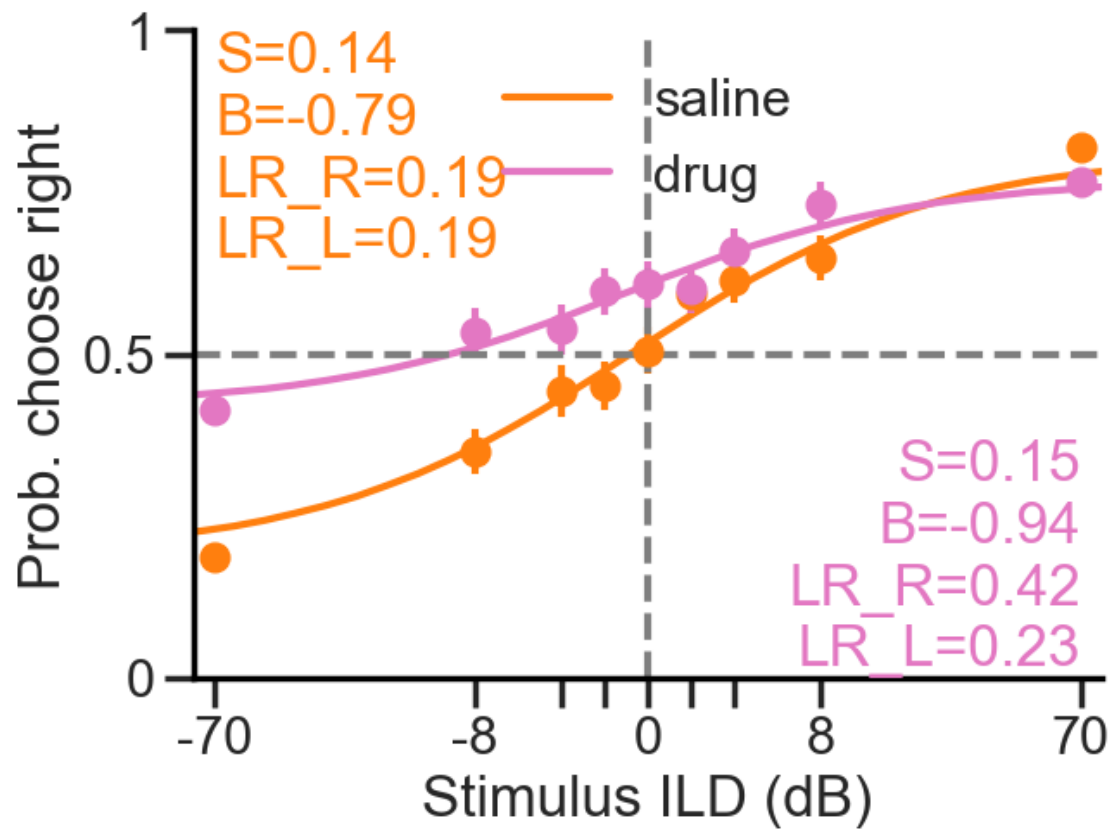


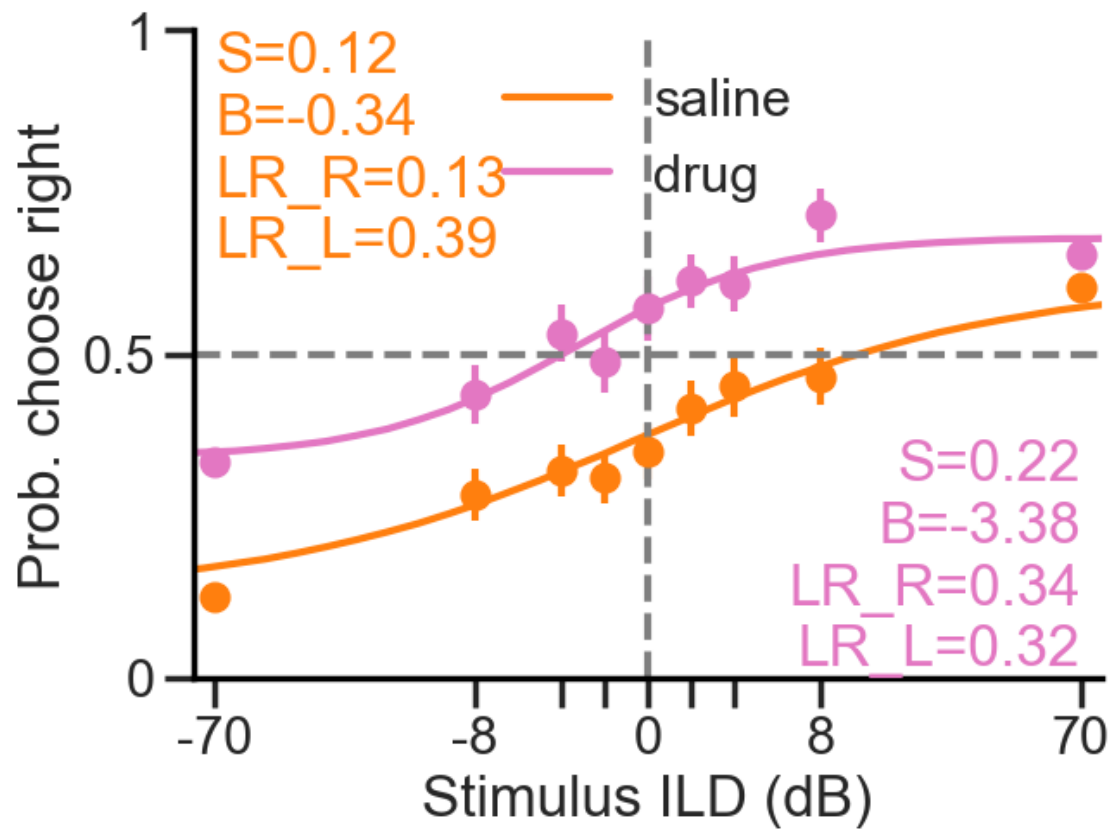


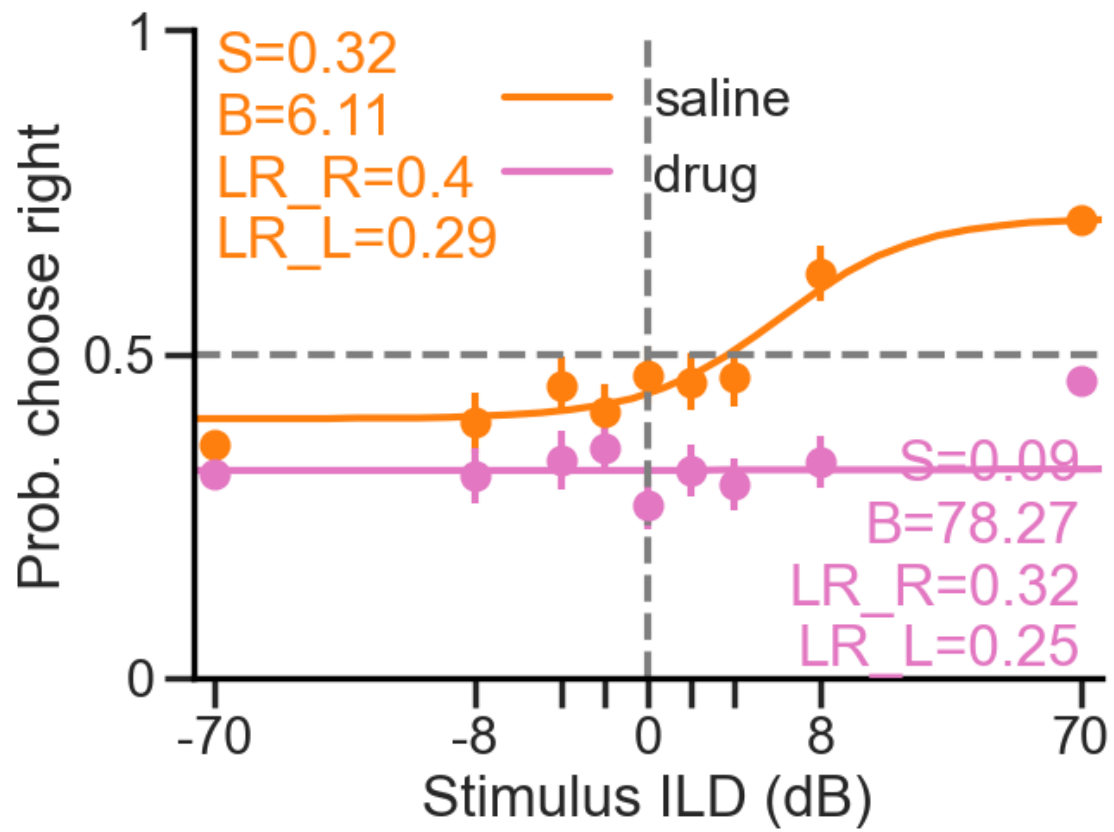


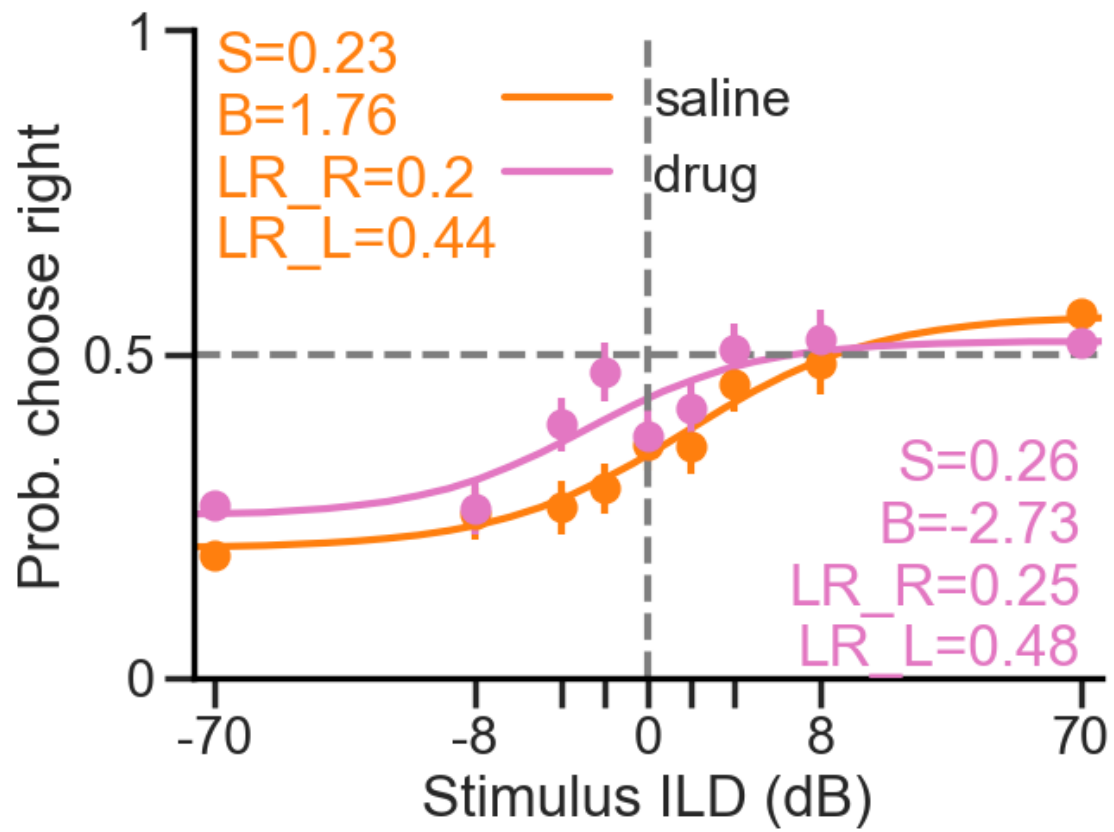


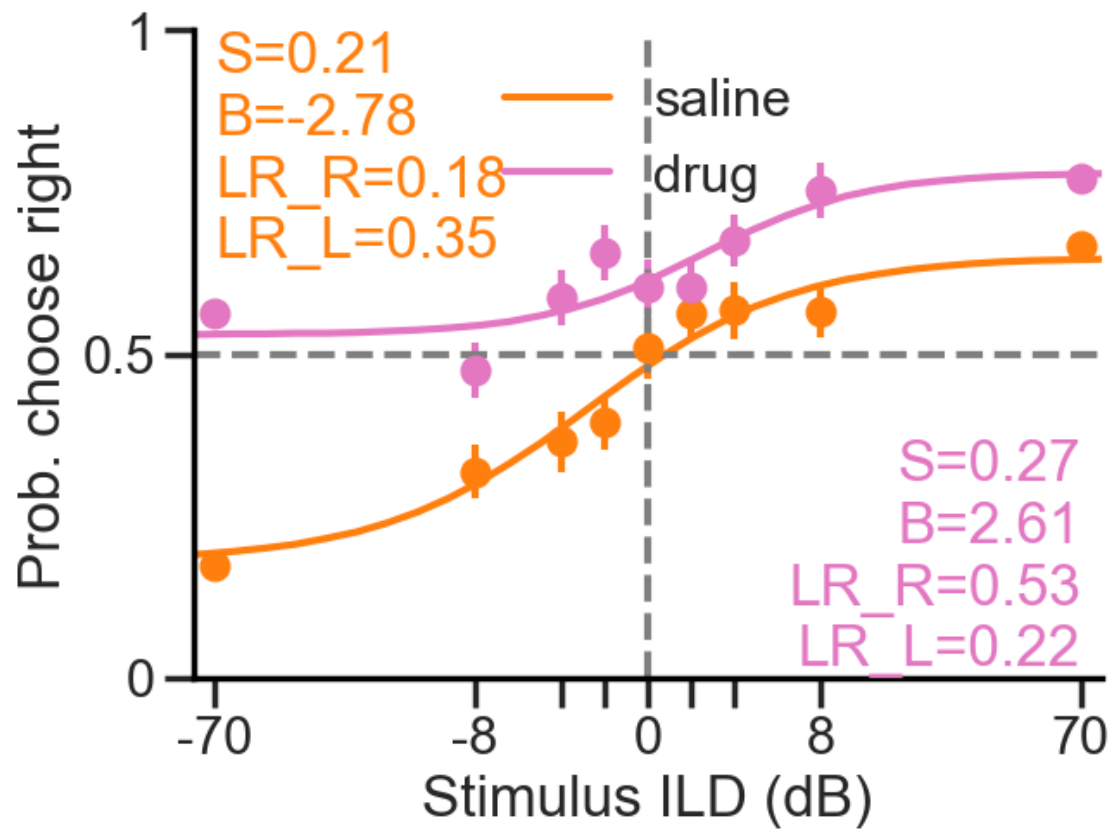




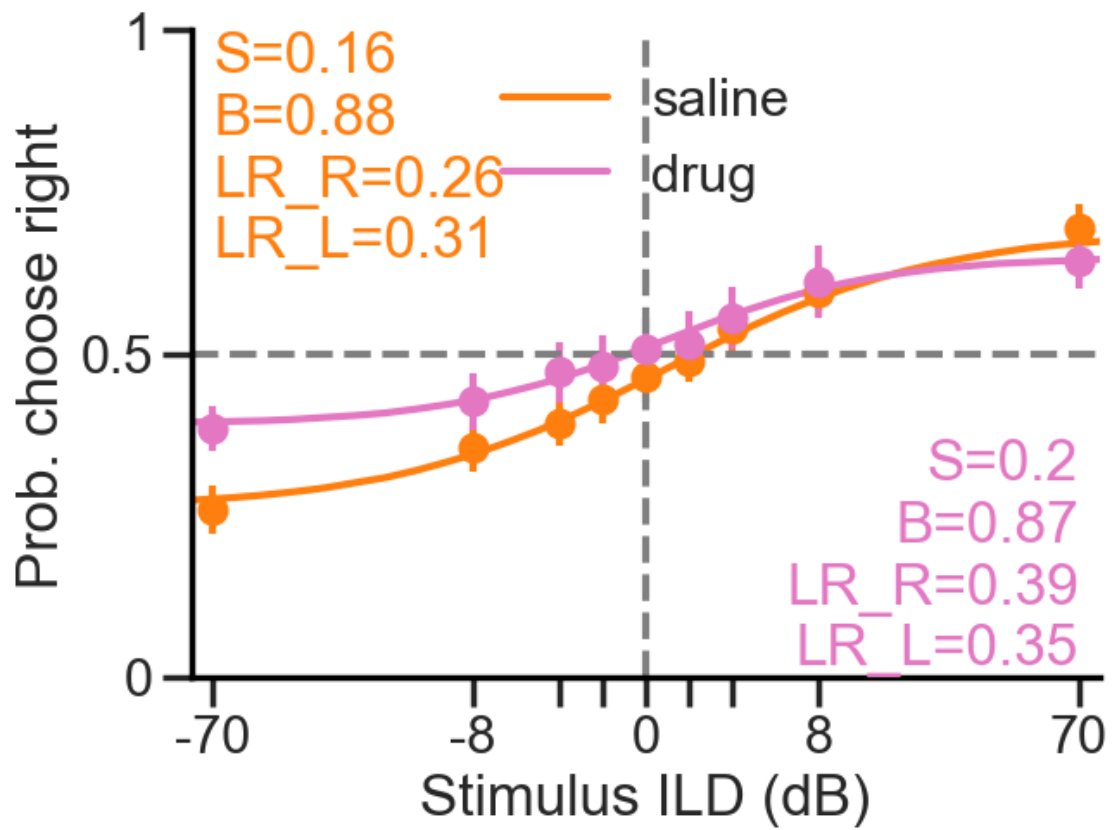






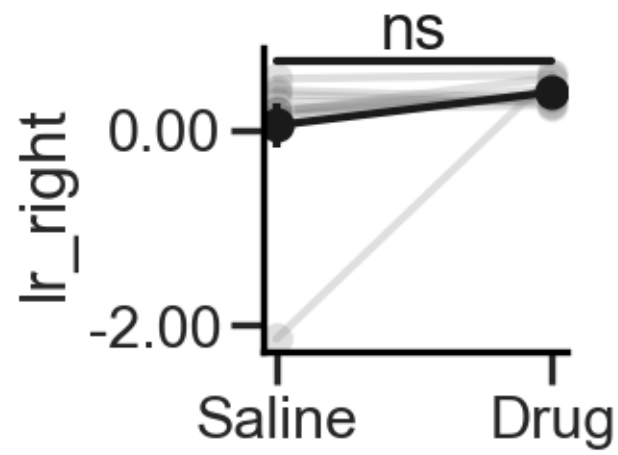
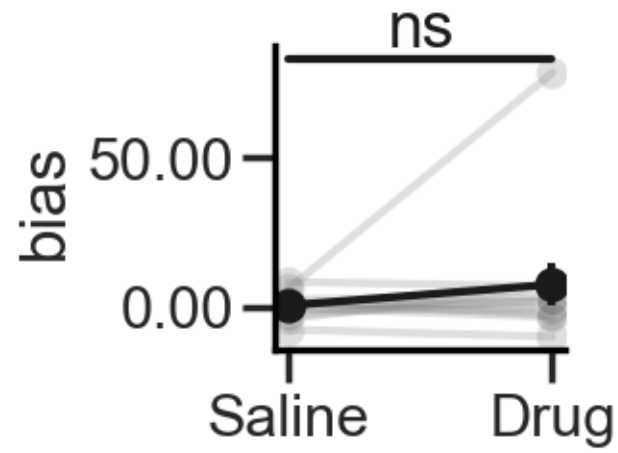
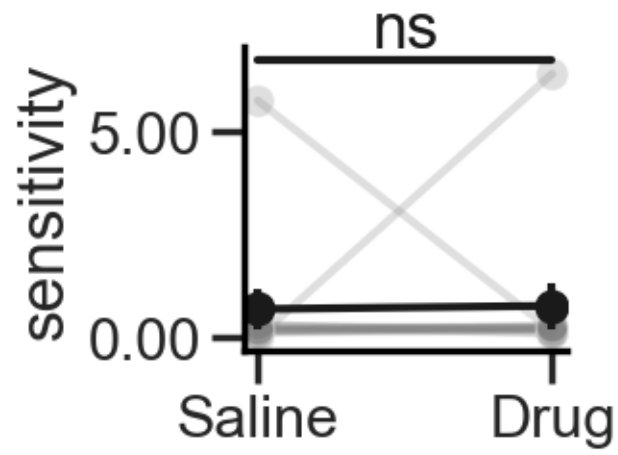


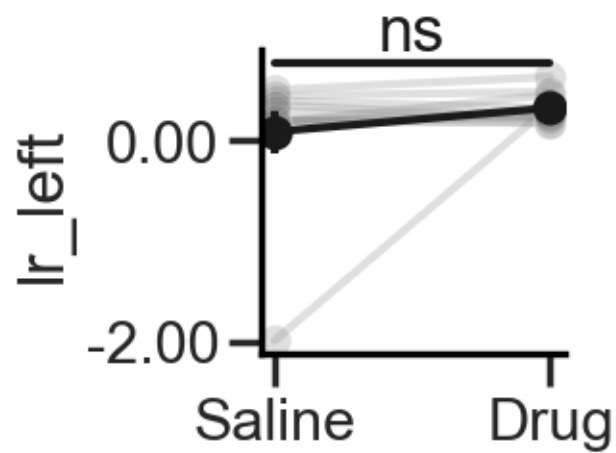
6.1.2 Mean across animals



Stats

sensitivity: t-statistic = -0.097, p-value = 0.925
bias: t-statistic = -1.053, p-value = 0.317
lr_right: t-statistic = -1.415, p-value = 0.188
lr_left: t-statistic = -1.111, p-value = 0.293

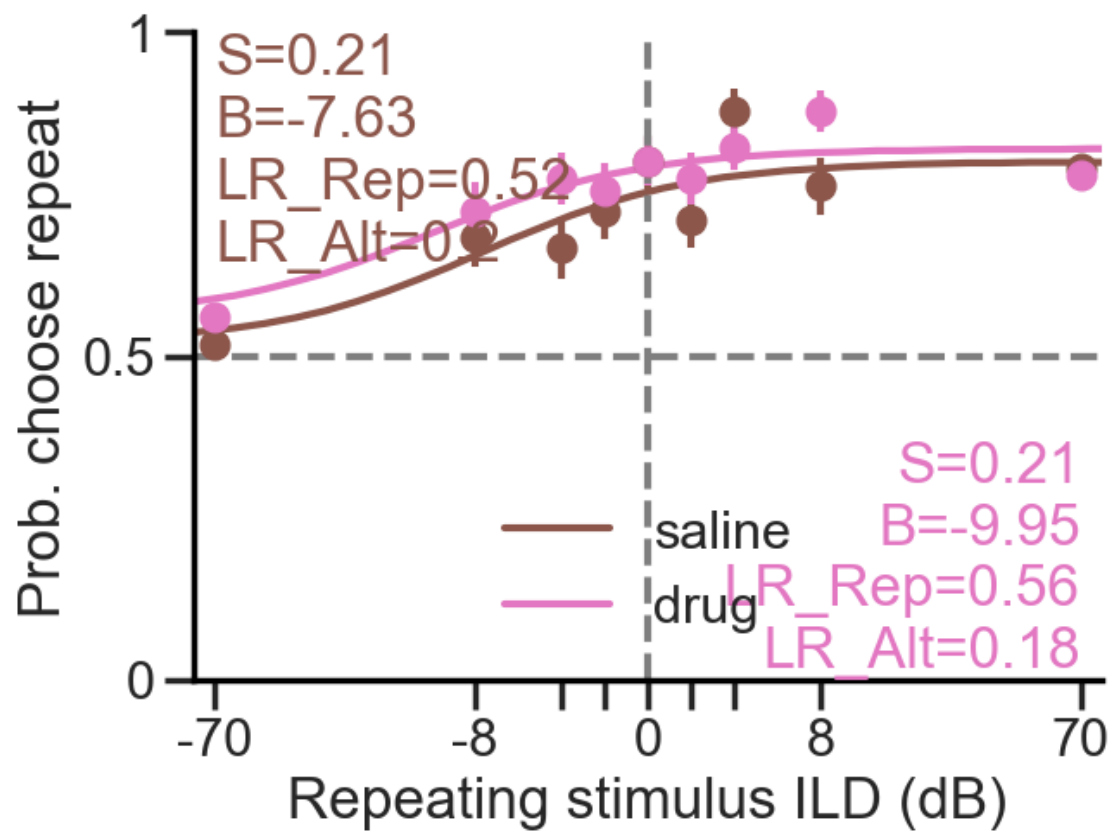


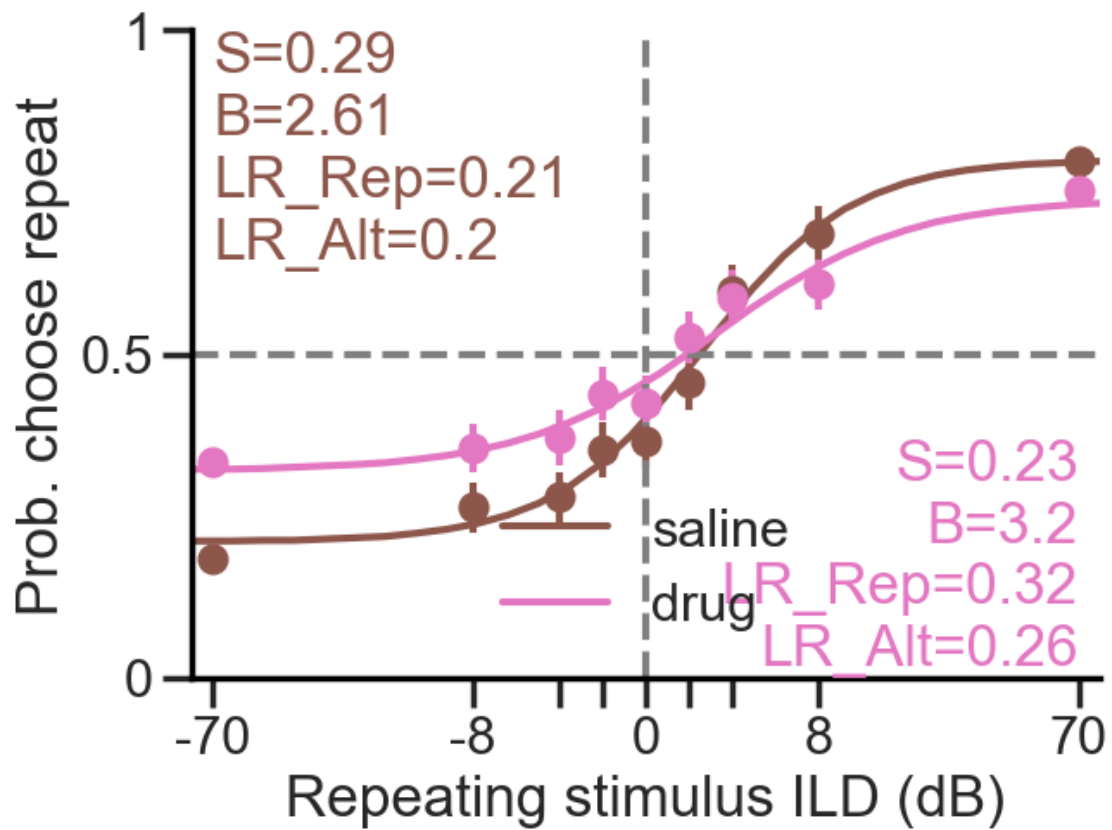


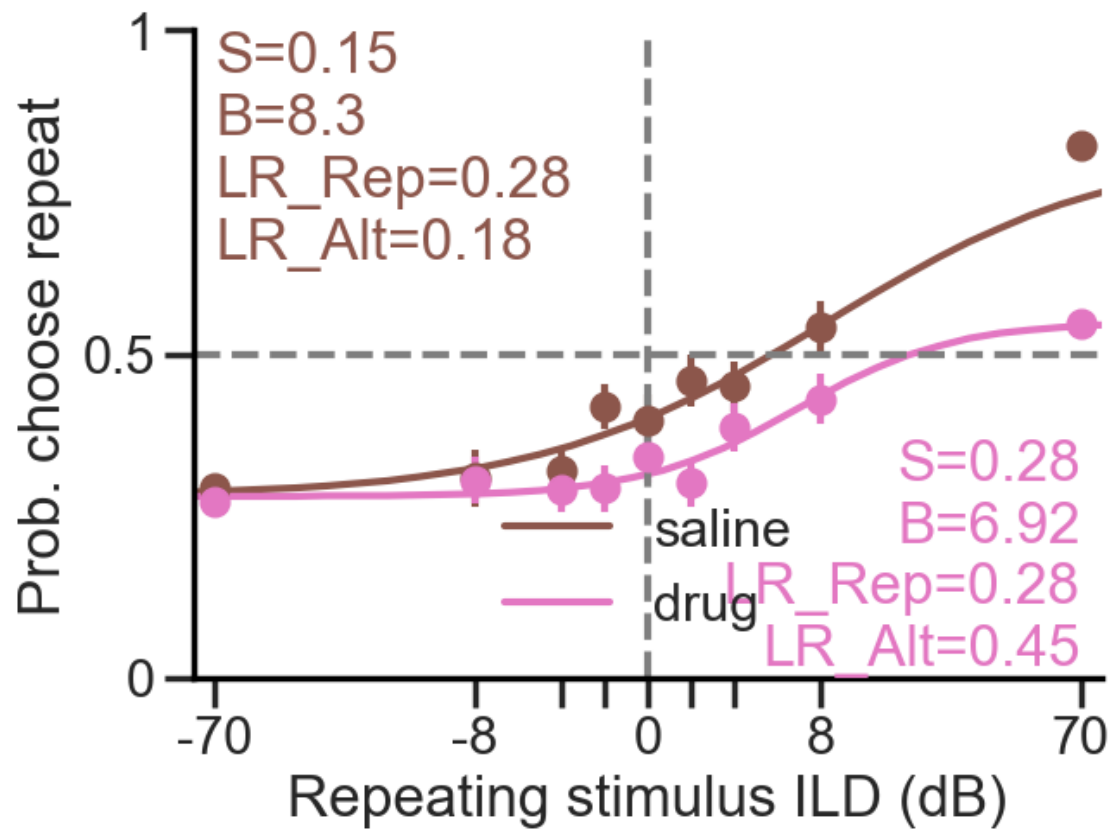
6.2 Probability choose REPEAT

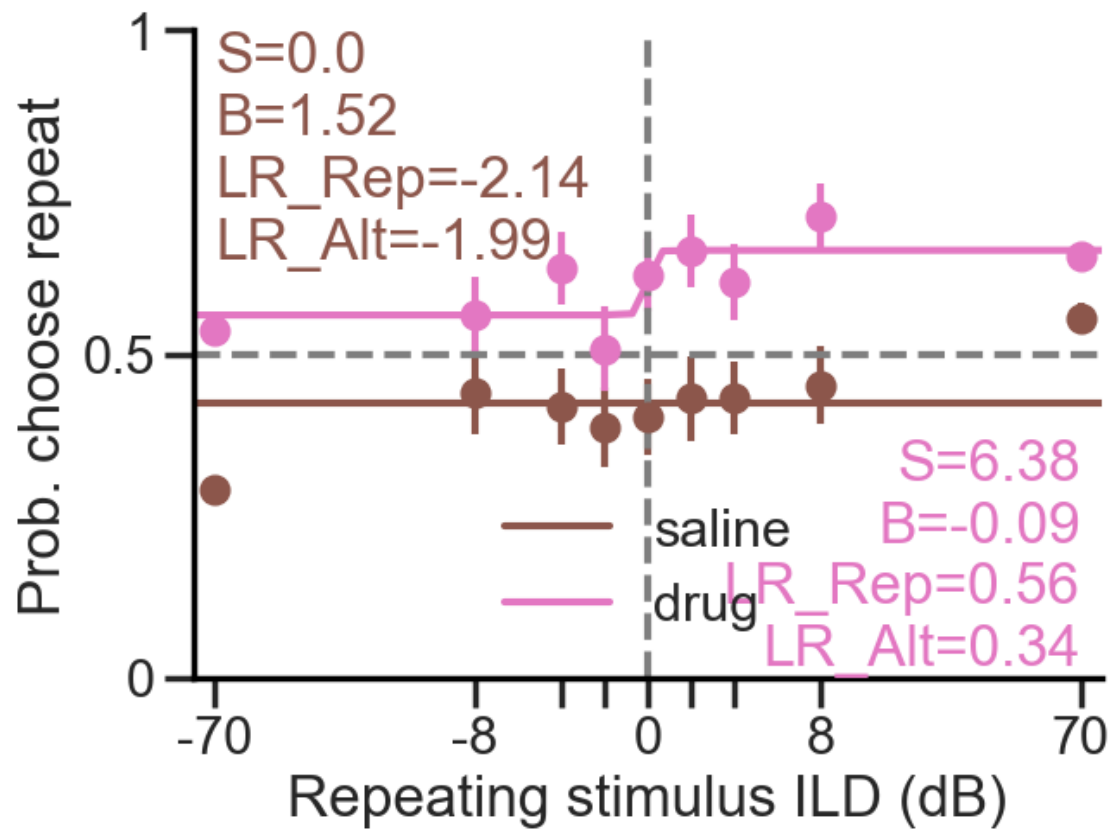
6.2.1 Single animals

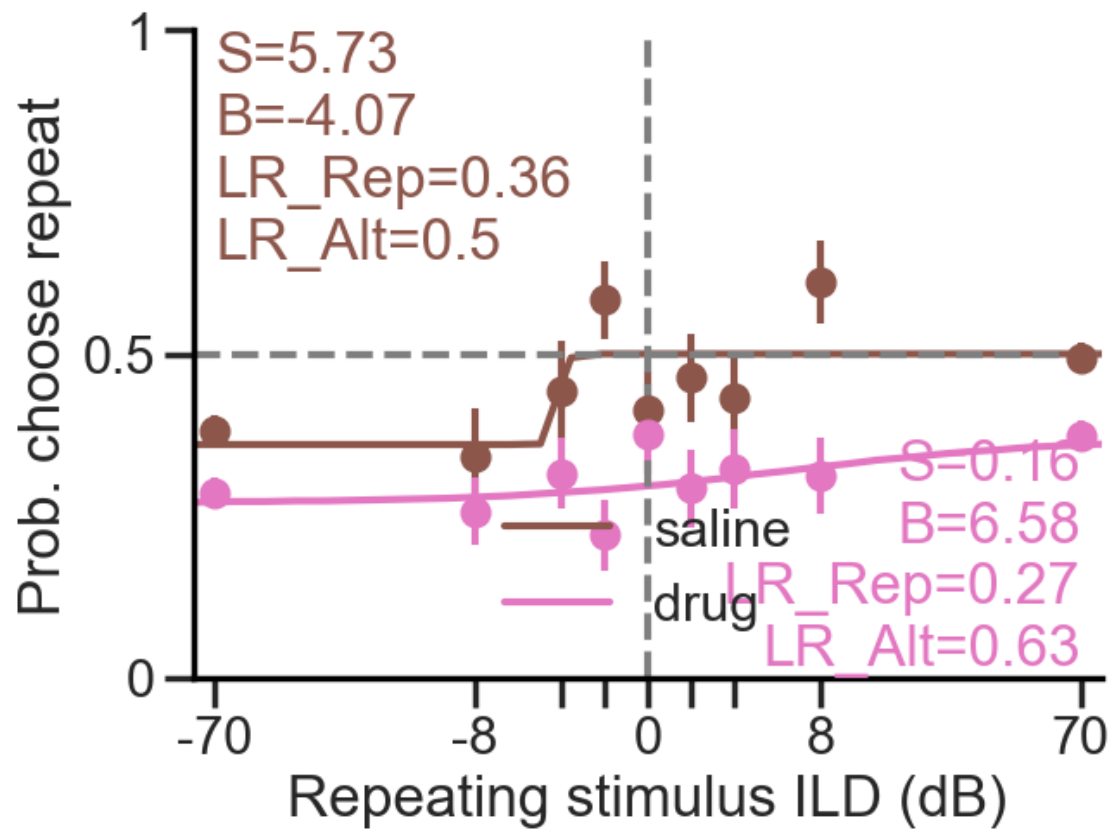
Animal 014
Animal 016
Animal 017
Animal 018
Animal 019
Animal 020
Animal 021
Animal 022
Animal 023
Animal 024
Animal 025

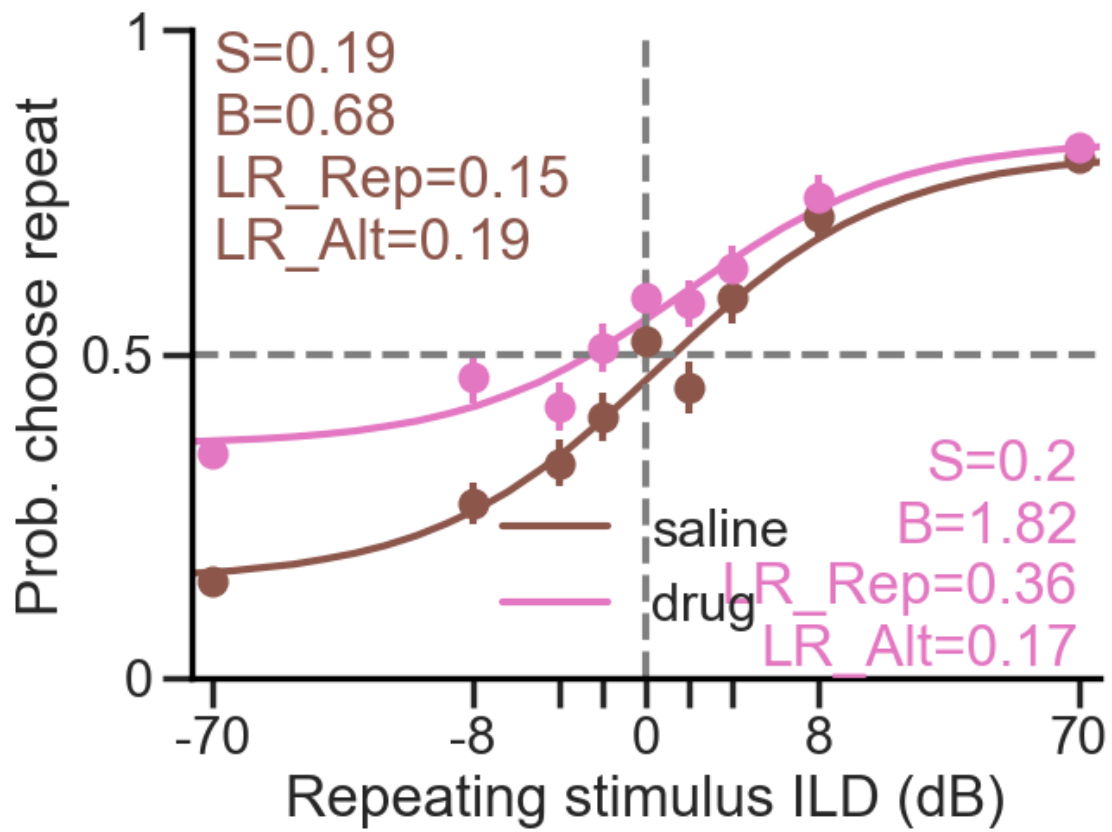


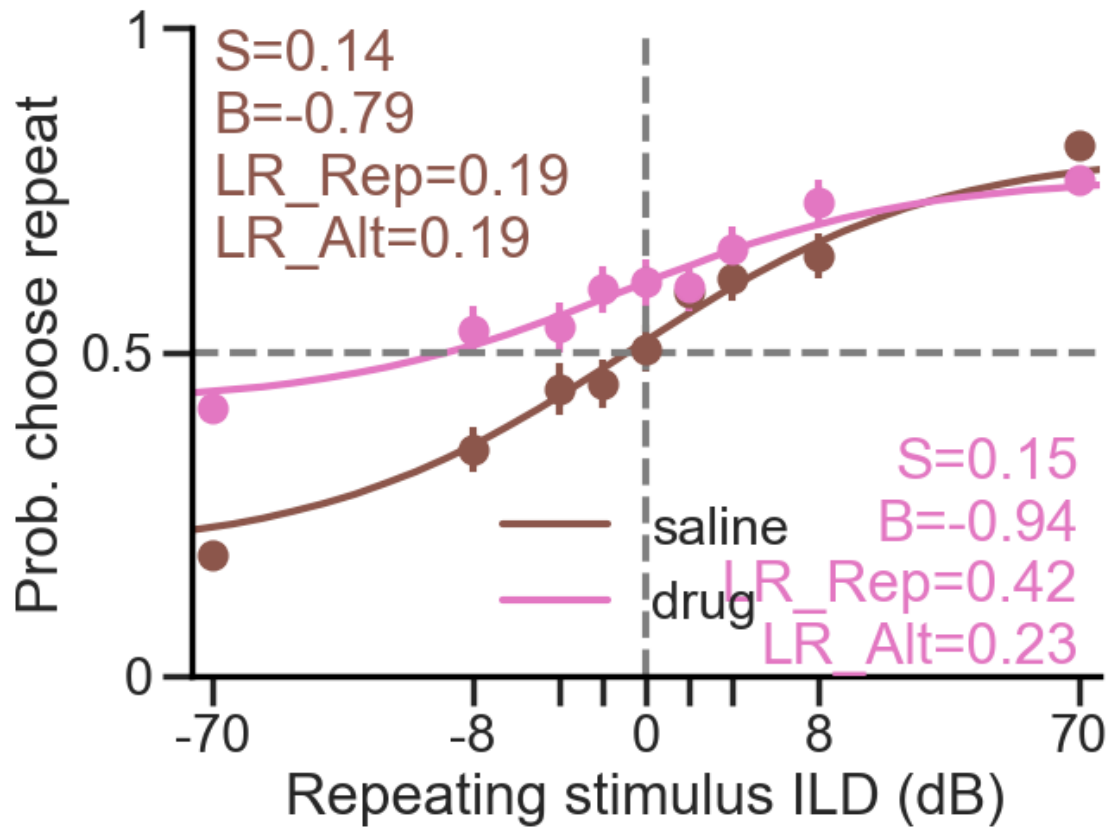


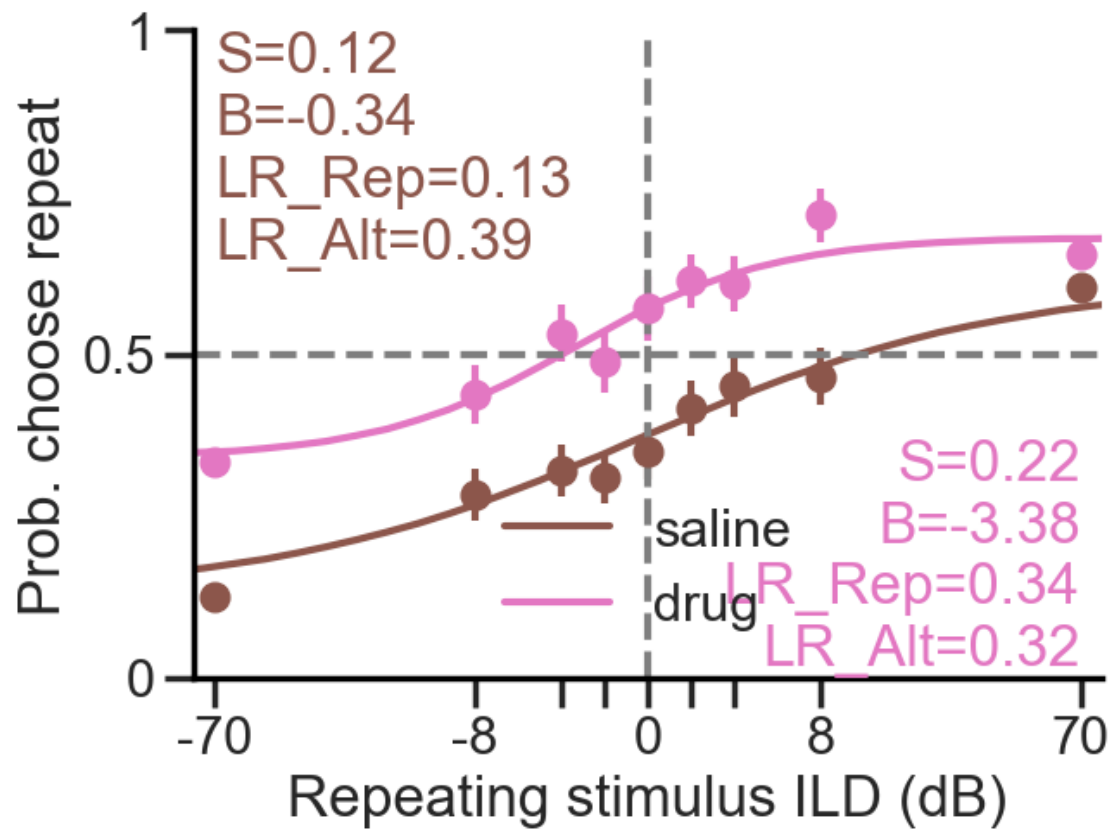


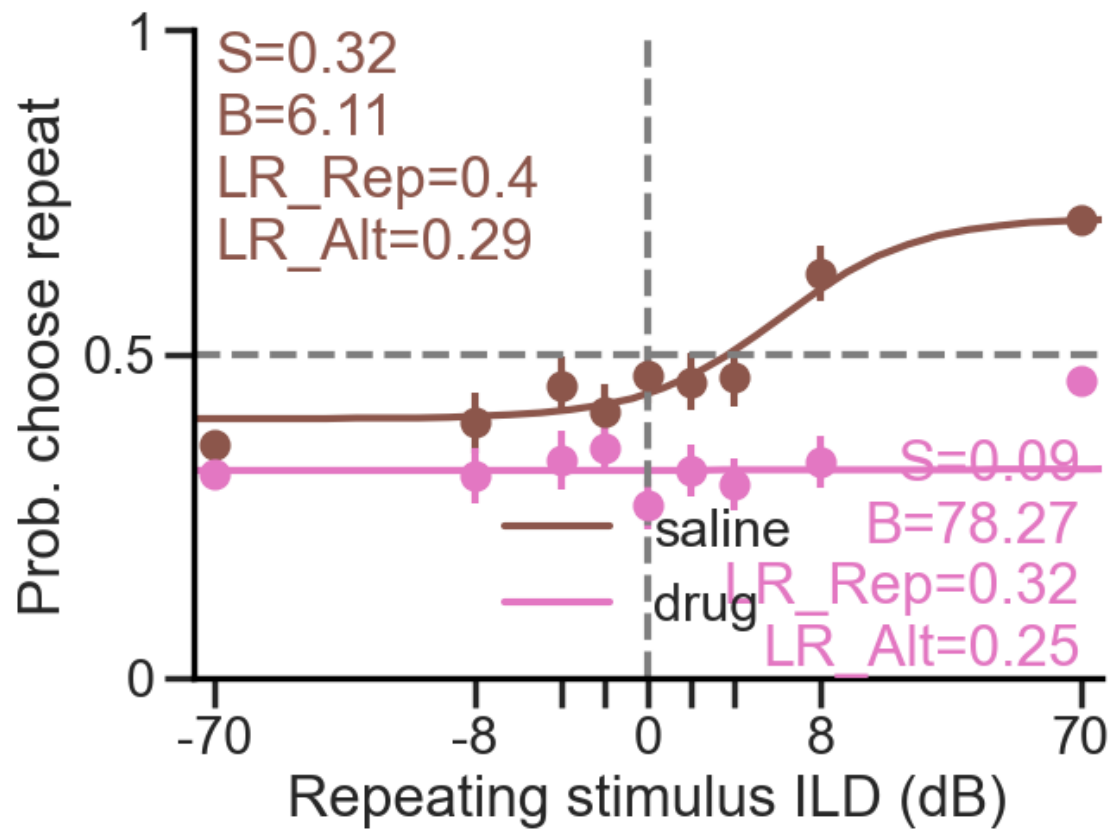


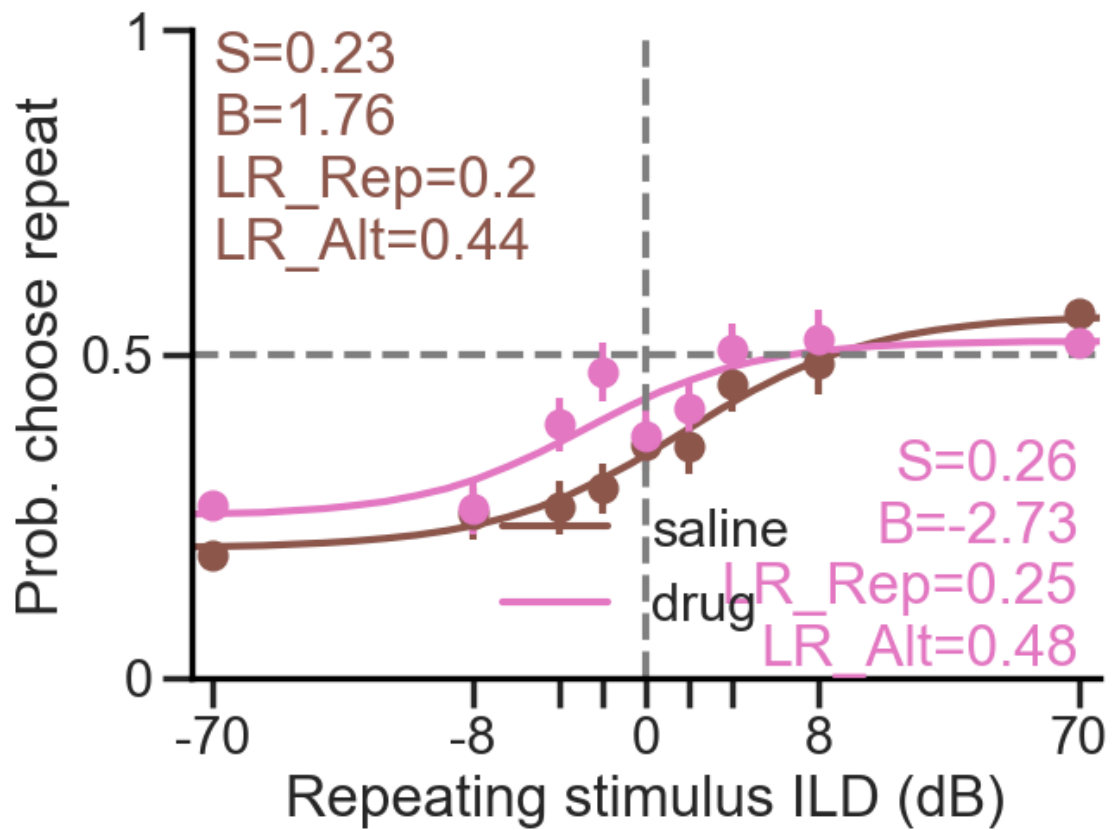


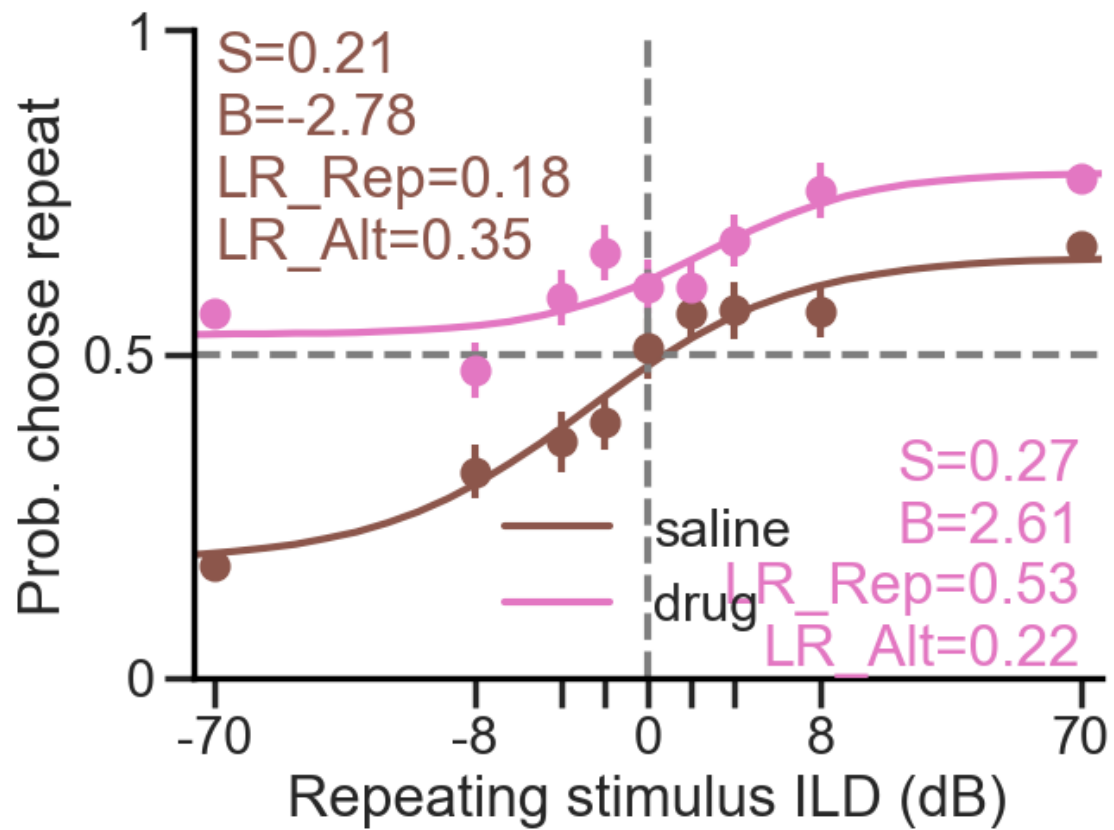




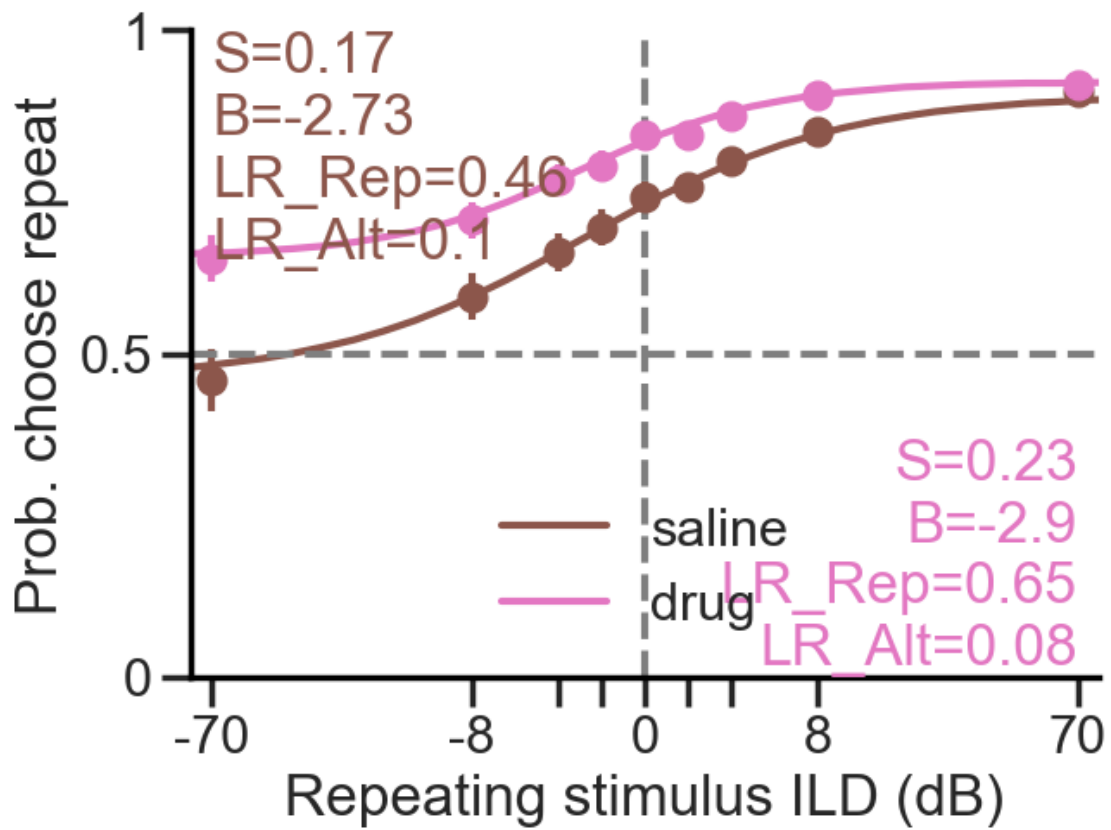






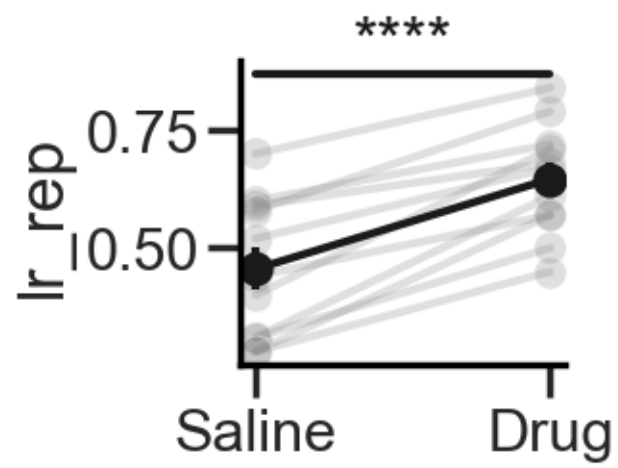
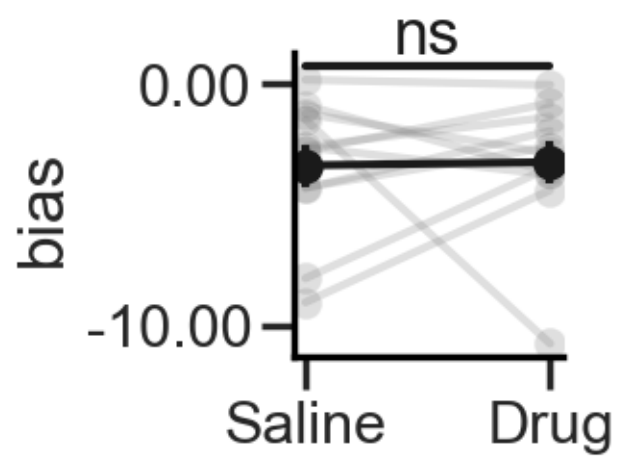
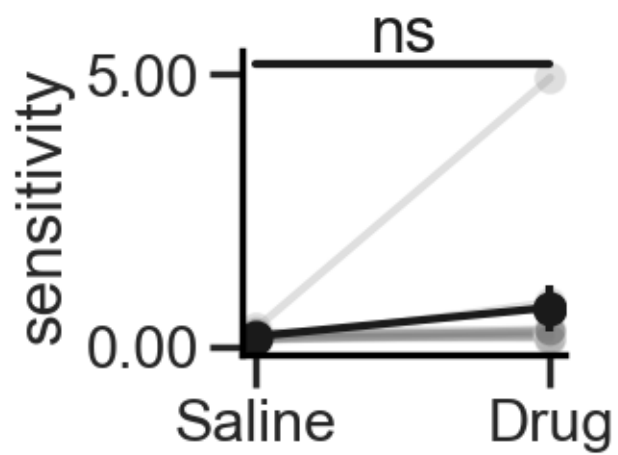


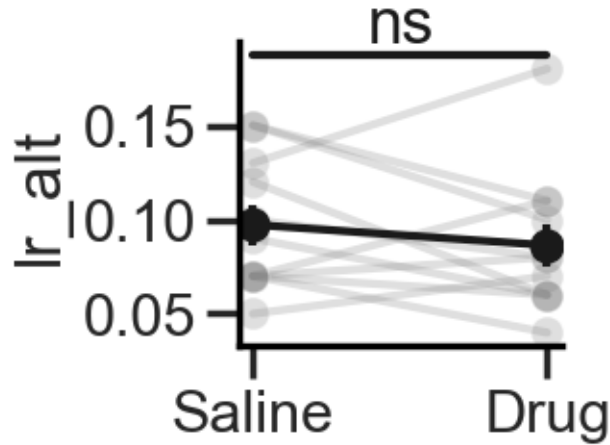
6.2.2 Mean across animals



Stats

sensitivity: t-statistic = -1.279, p-value = 0.230
bias: t-statistic = -0.110, p-value = 0.914
lr_rep: t-statistic = -8.182, p-value = 0.000
lr_alt: t-statistic = 0.992, p-value = 0.344





7 Kernels

7.1 Psychophysical kernels

7.1.1 Mean across animals

2AFC_6

C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6

Getting kernel of animal 014 (1/11)

get_pk took 1.80 seconds to run

Getting kernel of animal 016 (2/11)

get_pk took 2.12 seconds to run

Getting kernel of animal 017 (3/11)

get_pk took 2.17 seconds to run

Getting kernel of animal 018 (4/11)

get_pk took 1.23 seconds to run

Getting kernel of animal 019 (5/11)

get_pk took 1.33 seconds to run

Getting kernel of animal 020 (6/11)

get_pk took 2.22 seconds to run

Getting kernel of animal 021 (7/11)

get_pk took 2.02 seconds to run

Getting kernel of animal 022 (8/11)

get_pk took 1.96 seconds to run

Getting kernel of animal 023 (9/11)

get_pk took 1.68 seconds to run

Getting kernel of animal 024 (10/11)

get_pk took 1.84 seconds to run

Getting kernel of animal 025 (11/11)

get_pk took 1.68 seconds to run

```

get_mean_pk took 20.06 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/11)
get_pk took 1.74 seconds to run
Getting kernel of animal 016 (2/11)
get_pk took 1.97 seconds to run
Getting kernel of animal 017 (3/11)
get_pk took 2.02 seconds to run
Getting kernel of animal 018 (4/11)
get_pk took 1.16 seconds to run
Getting kernel of animal 019 (5/11)
get_pk took 1.25 seconds to run
Getting kernel of animal 020 (6/11)
get_pk took 2.10 seconds to run
Getting kernel of animal 021 (7/11)
get_pk took 1.95 seconds to run
Getting kernel of animal 022 (8/11)
get_pk took 1.93 seconds to run
Getting kernel of animal 023 (9/11)
get_pk took 1.68 seconds to run
Getting kernel of animal 024 (10/11)
get_pk took 1.82 seconds to run
Getting kernel of animal 025 (11/11)
get_pk took 1.68 seconds to run
get_mean_pk took 19.31 seconds to run
plot_drug_pk took 39.39 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/11)
get_pk took 1.78 seconds to run
Getting kernel of animal 016 (2/11)
get_pk took 1.93 seconds to run
Getting kernel of animal 017 (3/11)
get_pk took 1.99 seconds to run
Getting kernel of animal 018 (4/11)
get_pk took 1.13 seconds to run
Getting kernel of animal 019 (5/11)
get_pk took 1.22 seconds to run
Getting kernel of animal 020 (6/11)
get_pk took 2.12 seconds to run
Getting kernel of animal 021 (7/11)
get_pk took 1.88 seconds to run
Getting kernel of animal 022 (8/11)
get_pk took 1.91 seconds to run
Getting kernel of animal 023 (9/11)
get_pk took 1.68 seconds to run
Getting kernel of animal 024 (10/11)

```

```

get_pk took 1.87 seconds to run
Getting kernel of animal 025 (11/11)
get_pk took 1.67 seconds to run
get_mean_pk took 19.18 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/11)
get_pk took 1.77 seconds to run
Getting kernel of animal 016 (2/11)
get_pk took 1.90 seconds to run
Getting kernel of animal 017 (3/11)
get_pk took 1.98 seconds to run
Getting kernel of animal 018 (4/11)
get_pk took 1.16 seconds to run
Getting kernel of animal 019 (5/11)
get_pk took 1.23 seconds to run
Getting kernel of animal 020 (6/11)
get_pk took 2.07 seconds to run
Getting kernel of animal 021 (7/11)
get_pk took 1.90 seconds to run
Getting kernel of animal 022 (8/11)
get_pk took 1.86 seconds to run
Getting kernel of animal 023 (9/11)
get_pk took 1.68 seconds to run
Getting kernel of animal 024 (10/11)
get_pk took 1.88 seconds to run
Getting kernel of animal 025 (11/11)
get_pk took 1.71 seconds to run
get_mean_pk took 19.16 seconds to run
plot_drug_pk took 38.36 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/11)
get_pk took 1.69 seconds to run
Getting kernel of animal 016 (2/11)
get_pk took 1.90 seconds to run
Getting kernel of animal 017 (3/11)
get_pk took 2.01 seconds to run
Getting kernel of animal 018 (4/11)
get_pk took 1.14 seconds to run
Getting kernel of animal 019 (5/11)
get_pk took 1.25 seconds to run
Getting kernel of animal 020 (6/11)
get_pk took 2.08 seconds to run
Getting kernel of animal 021 (7/11)
get_pk took 1.87 seconds to run
Getting kernel of animal 022 (8/11)
get_pk took 1.93 seconds to run

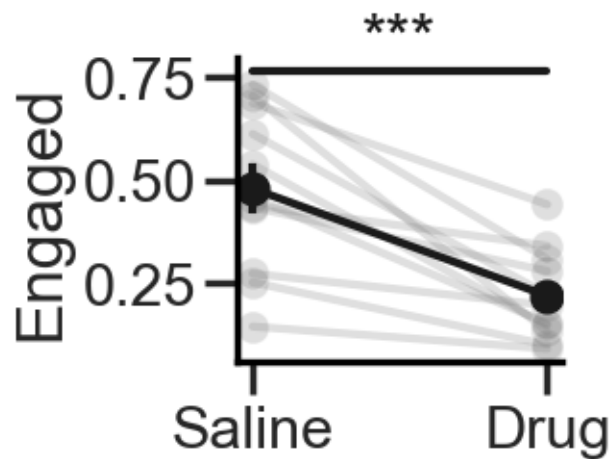
```

```
Getting kernel of animal 023 (9/11)
get_pk took 1.74 seconds to run
Getting kernel of animal 024 (10/11)
get_pk took 1.78 seconds to run
Getting kernel of animal 025 (11/11)
get_pk took 1.67 seconds to run
get_mean_pk took 19.06 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/11)
get_pk took 1.69 seconds to run
Getting kernel of animal 016 (2/11)
```

7.2 History kernels

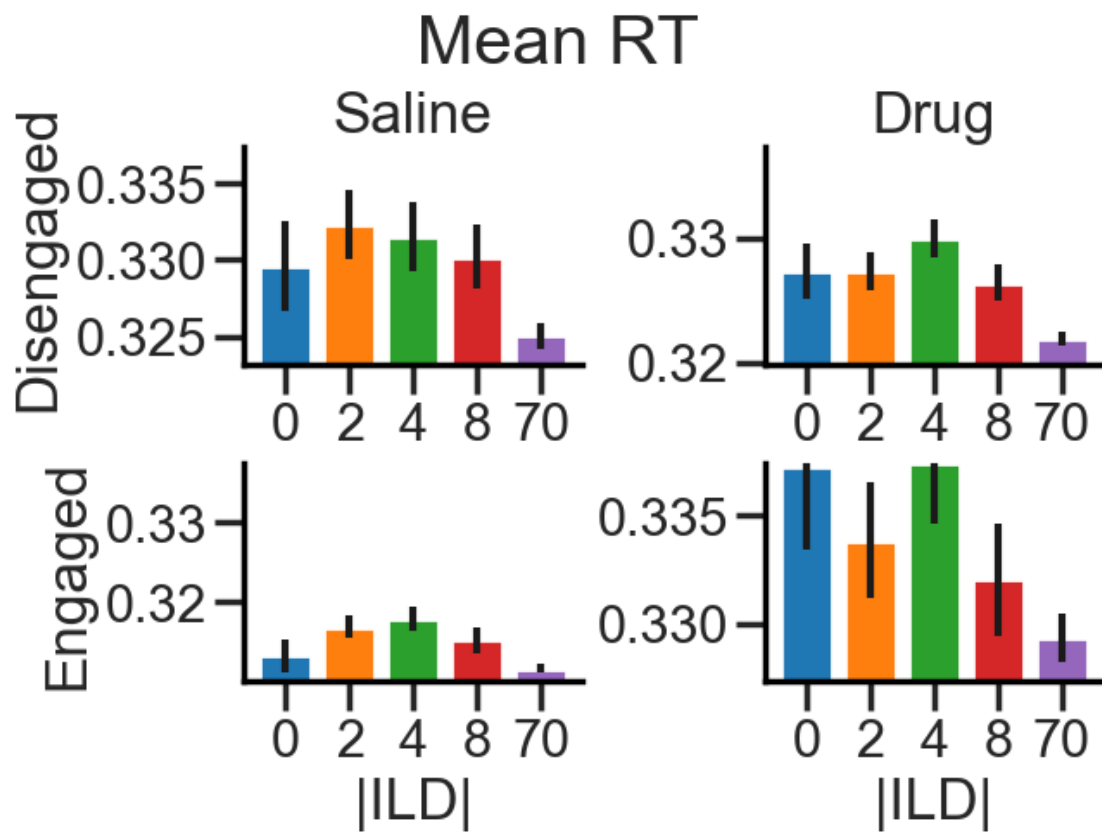
8 GLM HMM

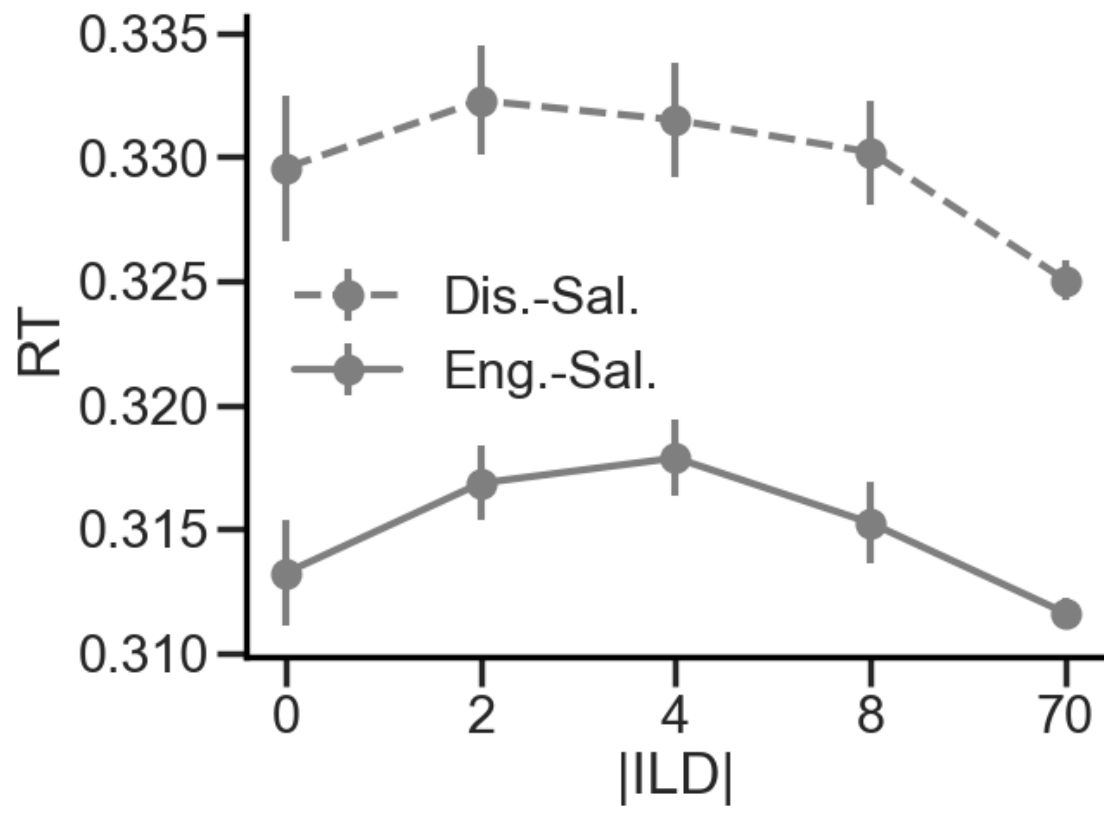
Engaged: t-statistic = 5.111, p-value = 0.000

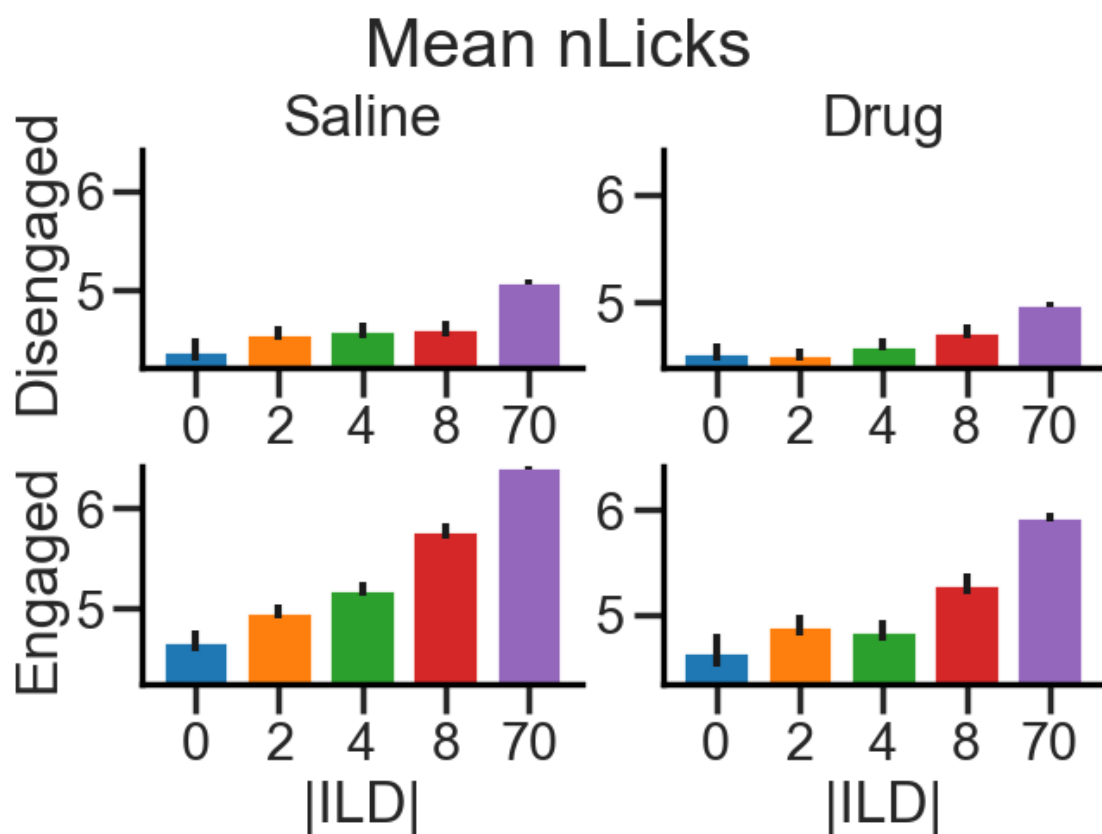


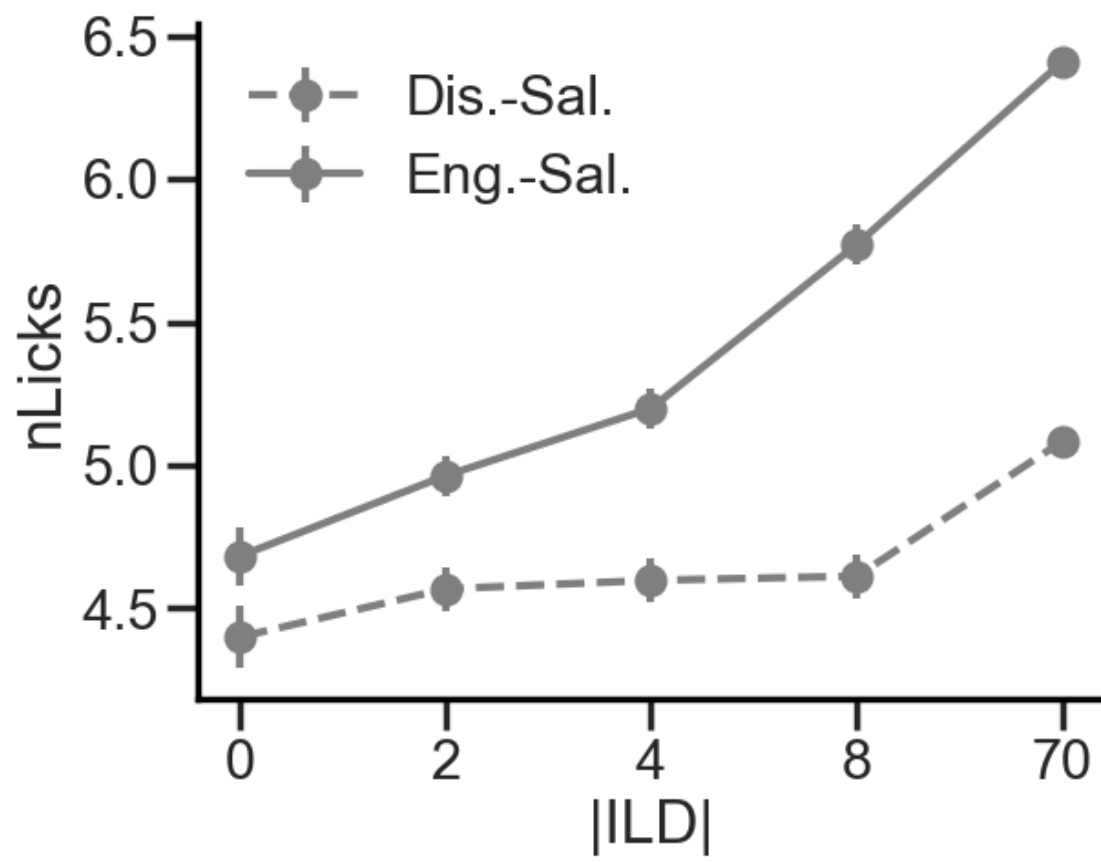
8.1 Licks

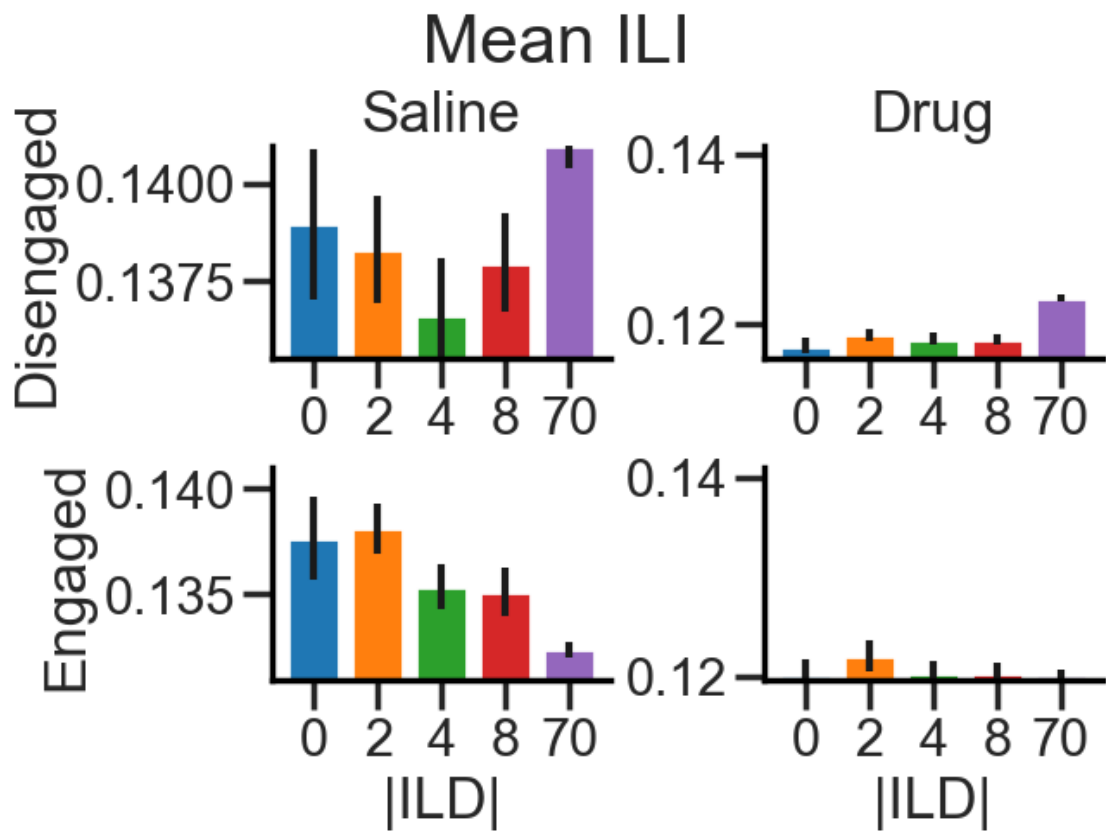
8.1.1 RTs

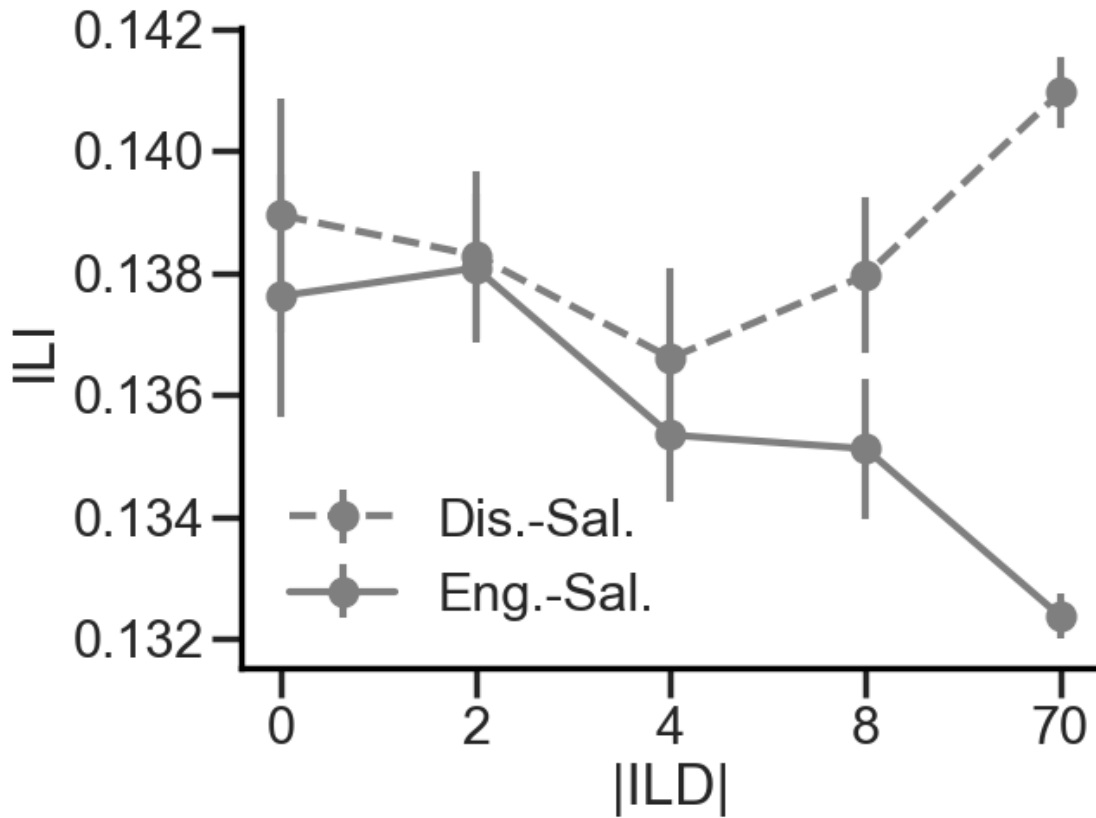












9 Serial dependence

Serial dependence requires a continuous readout of behavioral responses. As choices are binary (left/right), other readouts are needed. Let's try with reaction times (RT) and lick rate (nLicks)

	Trial	Side	PreviousSide	RepSide	RepTrial	Reward	Punish	Miss	\
0	153	1	NaN	0	1.0	1	0	0	
1	144	1	1.0	1	1.0	1	0	0	
2	146	1	0.0	0	1.0	1	0	0	
3	147	0	1.0	0	0.0	1	0	0	
4	148	1	0.0	0	0.0	1	0	0	

	WrongLick	Hit	...	nLicksLeft	nLicksRight	nLicks	leftILI	rightILI	\
0	0	1.0	...	0	8	1.360474	NaN	0.125586	
1	0	1.0	...	0	10	3.360474	NaN	0.106844	
2	0	1.0	...	0	7	0.360474	NaN	0.120250	
3	0	1.0	...	7	0	0.360474	0.15045	NaN	
4	0	1.0	...	0	10	3.360474	NaN	0.106567	

ILI	RT	RT2	SessionHalf	Engaged
-----	----	-----	-------------	---------

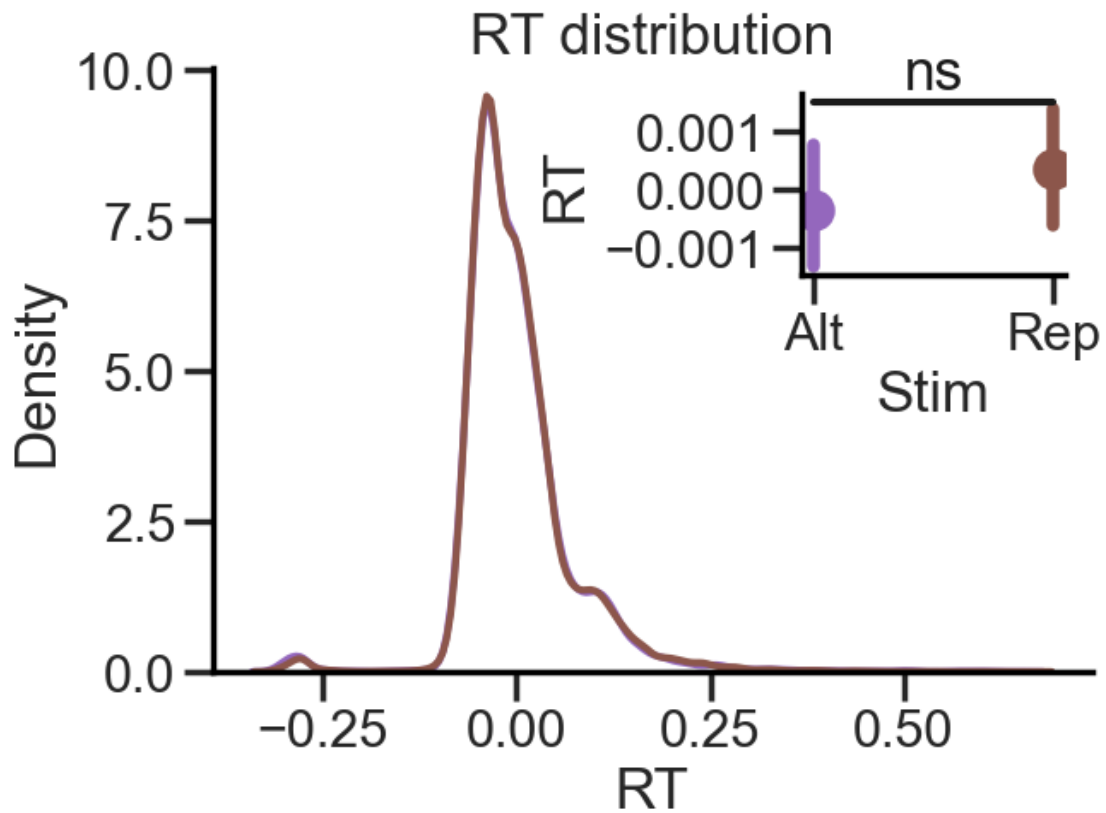
0	0.125586	0.088068	0.088084	1	1
1	0.106844	0.023468	0.023484	1	1
2	0.120250	0.112868	0.112884	1	1
3	0.150450	-0.011932	-0.011916	1	1
4	0.106567	-0.021532	-0.021516	1	1

[5 rows x 103 columns]

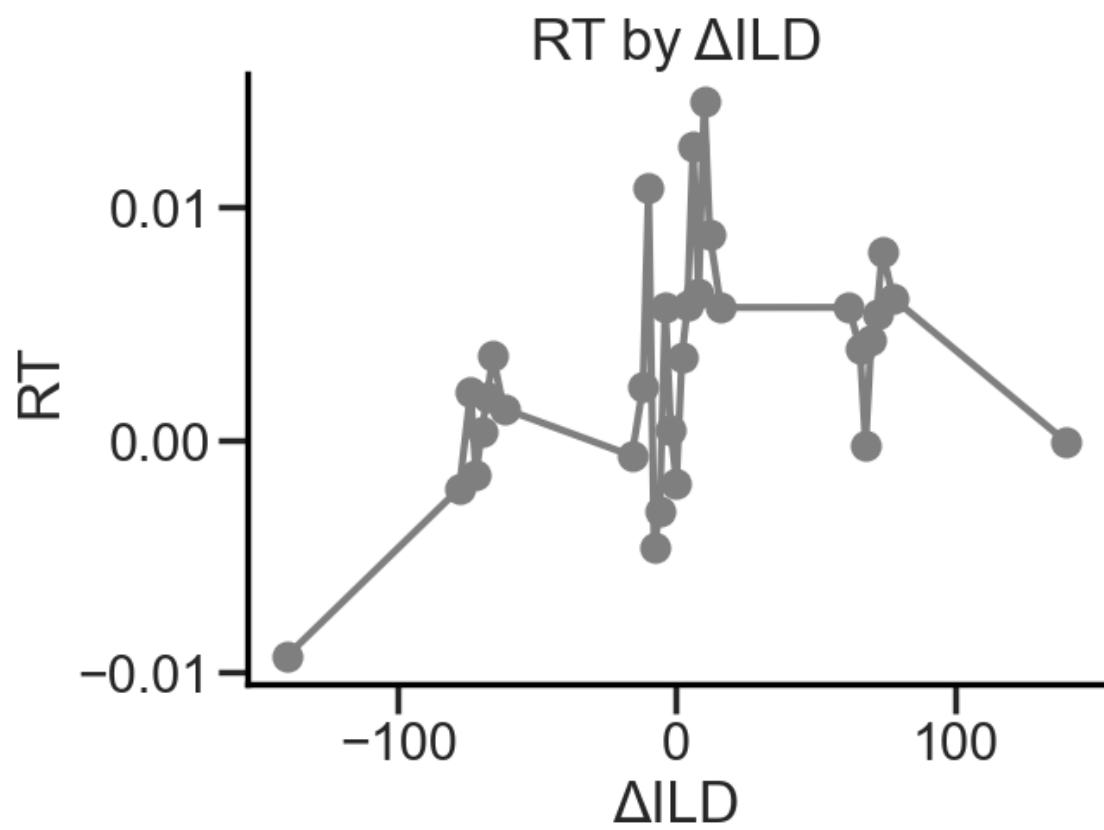
```
array([ -78.,  nan, -140.,  -74.,  -72.,  -70.,  -68.,  -66.,  -62.,
        -16.,  -12.,  -10.,   -8.,   -6.,   -4.,   -2.,   0.,   2.,
         4.,   6.,   8.,  10.,  12.,  16.,  62.,  66.,  68.,
        70.,  72.,  74.,  78., 140.]
```

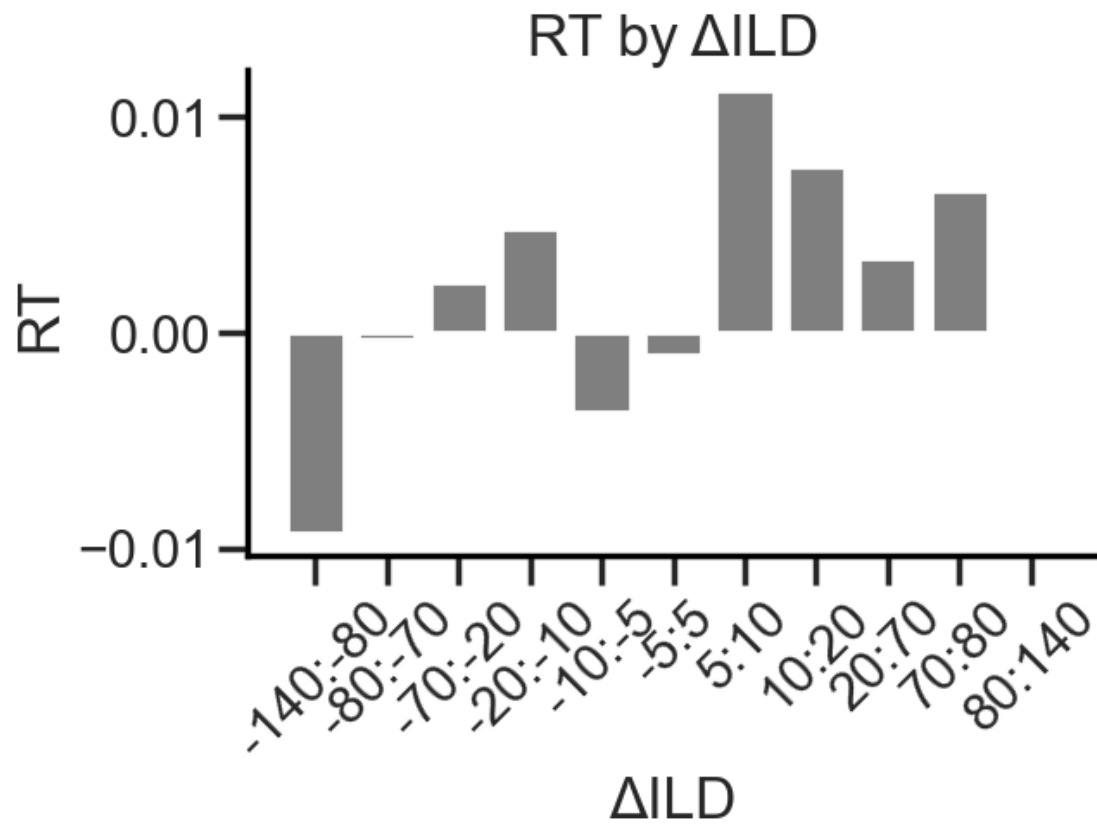
9.0.1 Measured as Reaction Times (RTs)

RT: t-statistic = -0.978, p-value = 0.328

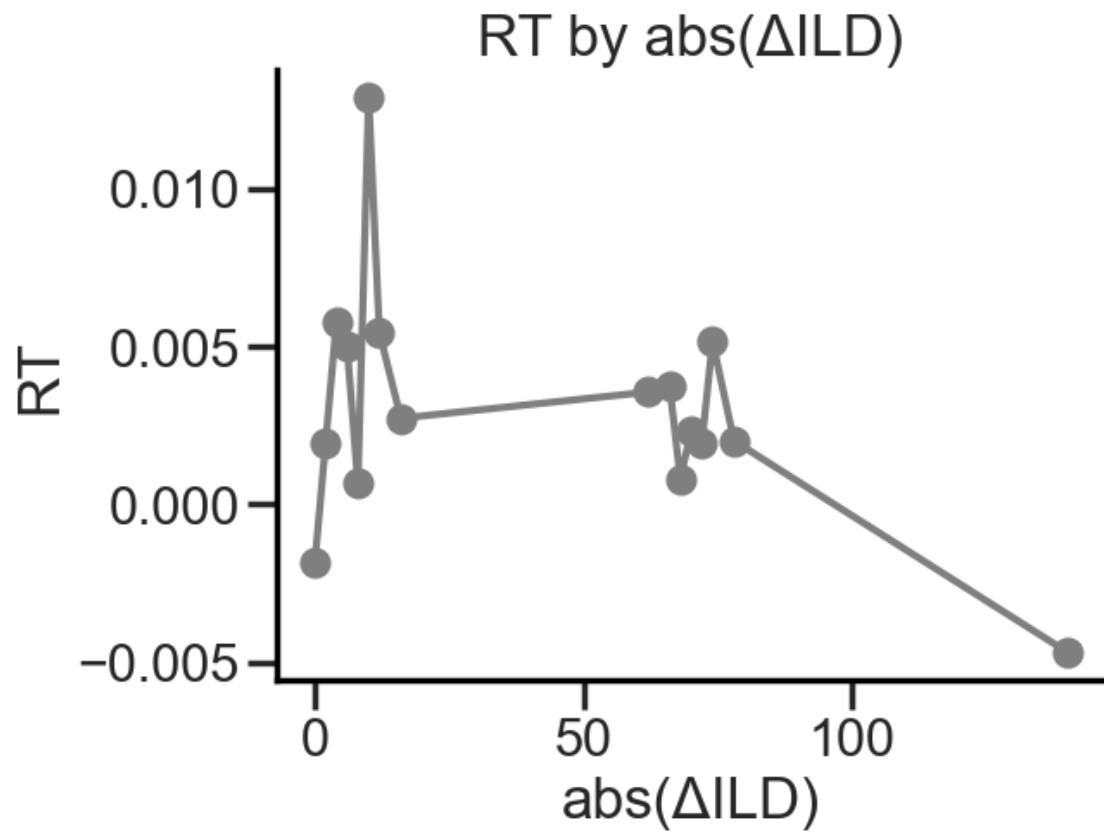


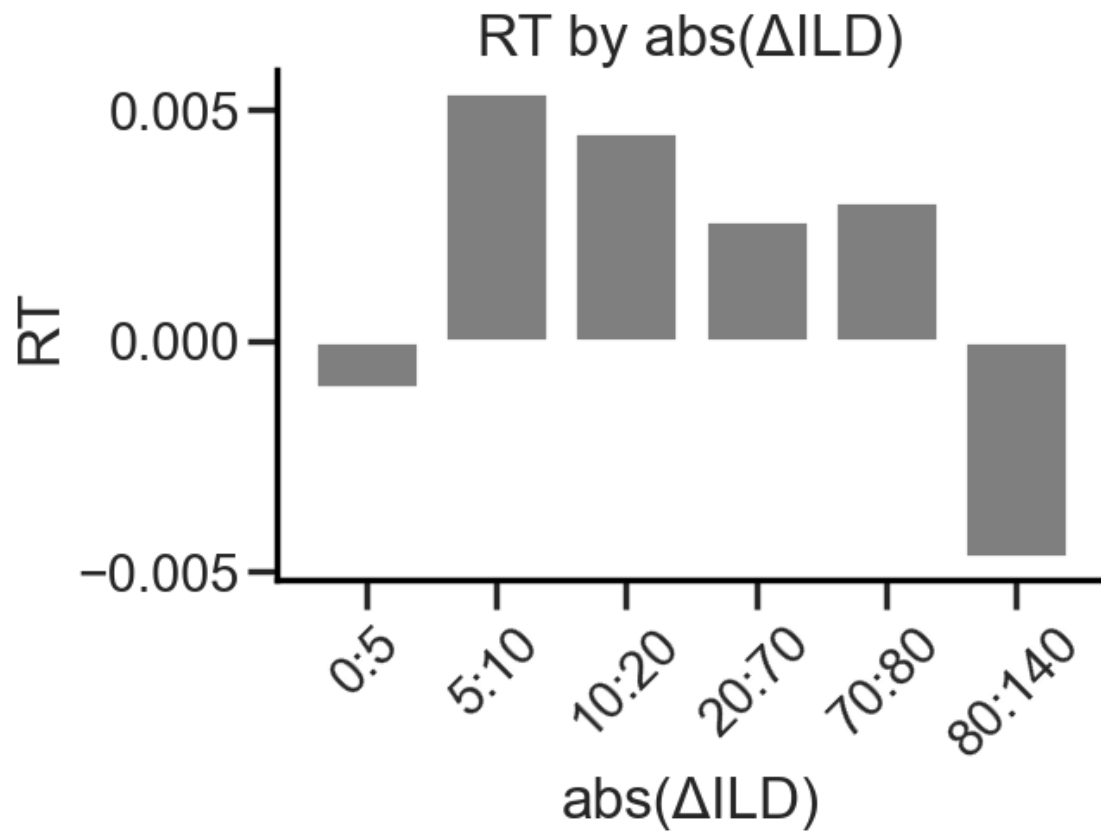
Δ ILD





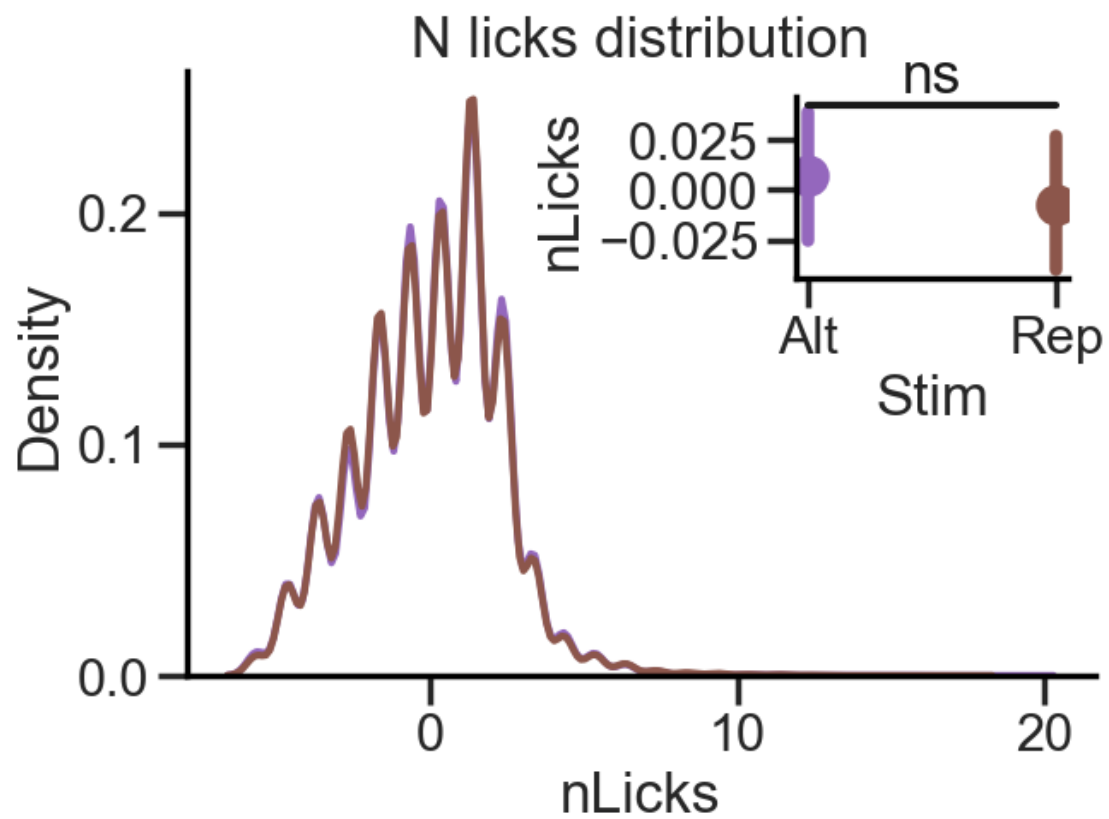
Absolute Δ ILD



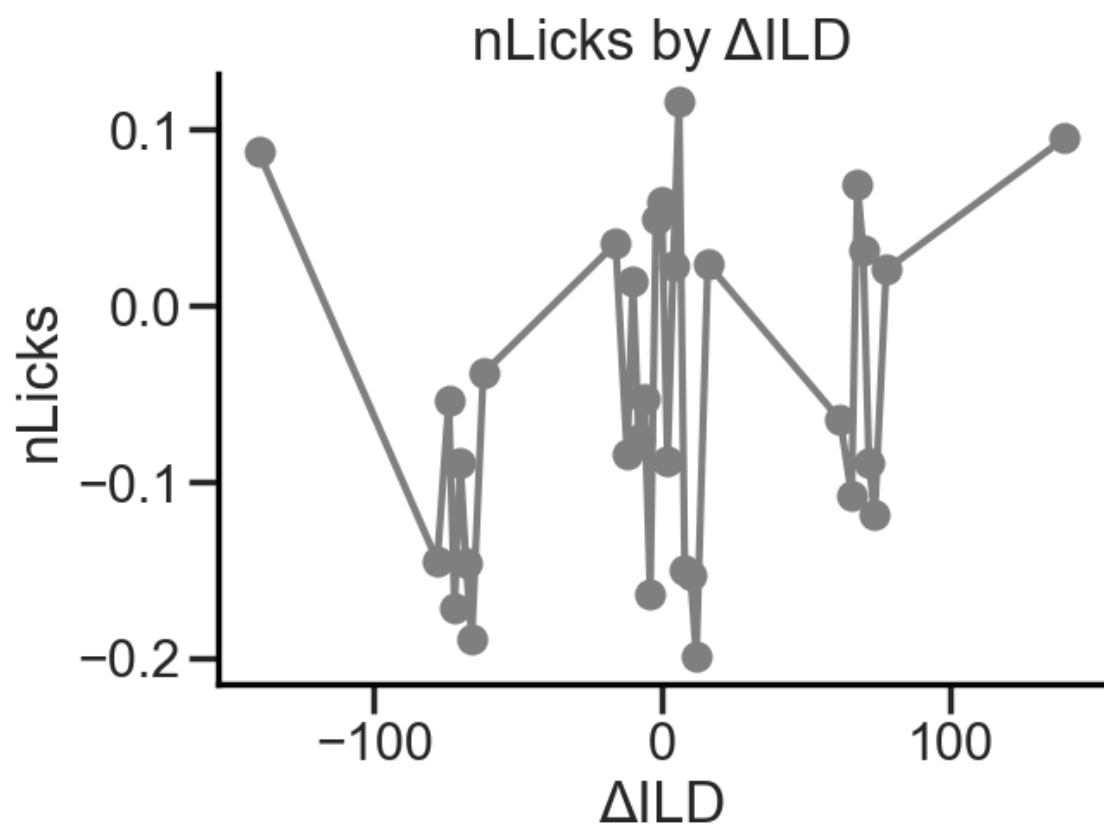


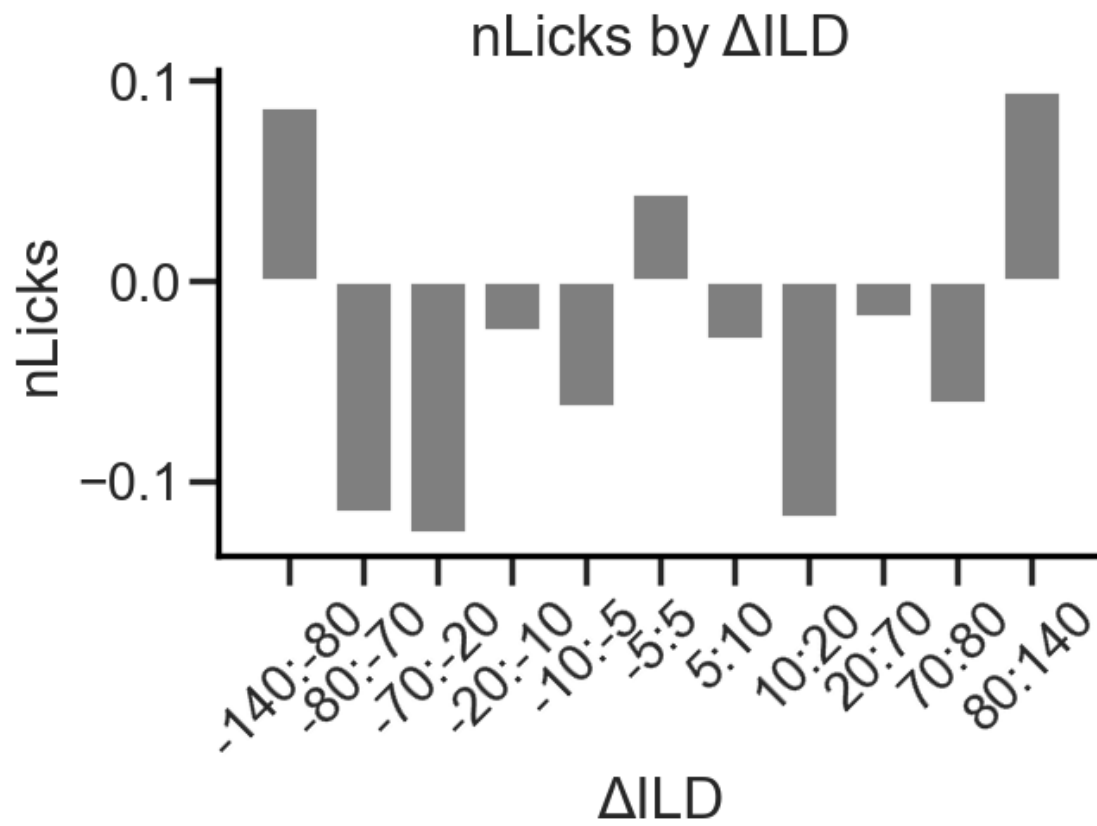
9.0.2 Measured as N licks

nLicks: t-statistic = 0.606, p-value = 0.545

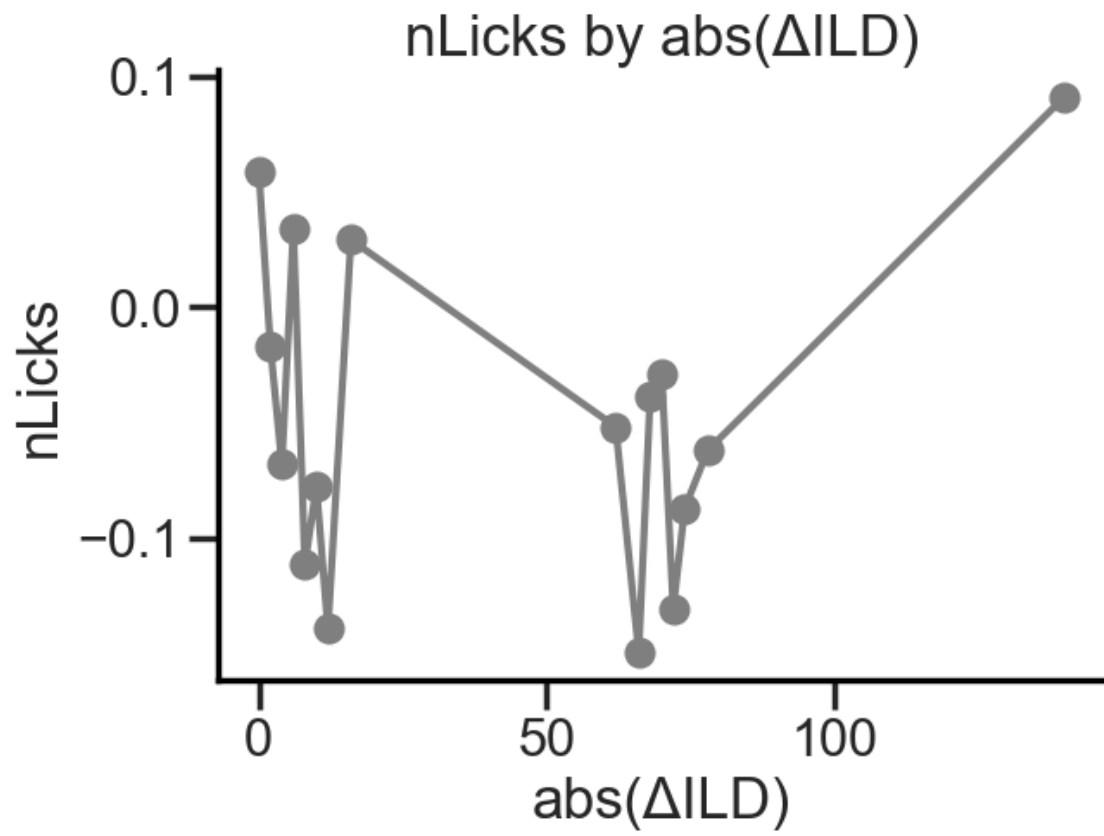


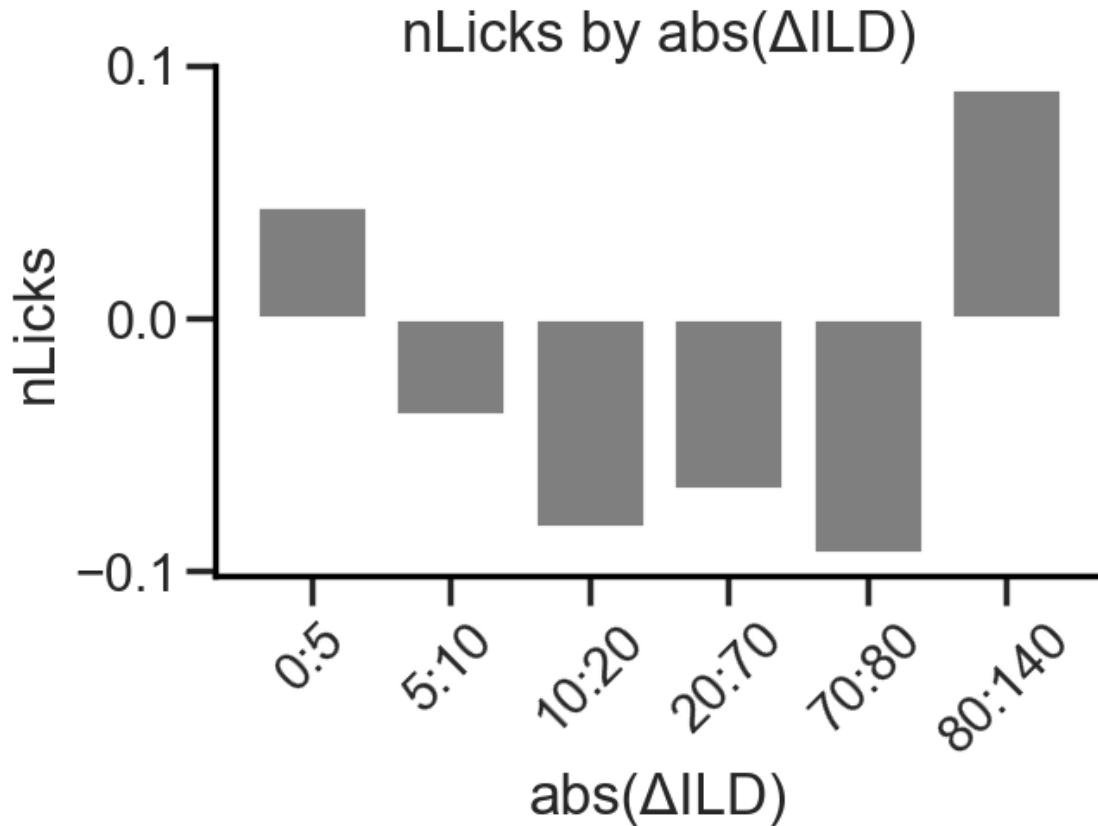
Δ ILD





Absolute Δ ILD





10 Save all notebook's figures

```
[NbConvertApp] Converting notebook notebook.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 70 image(s).
[NbConvertApp] Writing 4114424 bytes to notebook.html

Saved 70 images to notebook_files/

[NbConvertApp] Converting notebook notebook.ipynb to pdf
[NbConvertApp] Support files will be in notebook_files\
[NbConvertApp] Making directory .\notebook_files
[NbConvertApp] Writing 59309 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 2537560 bytes to notebook.pdf
```